



## **Final Year Project Report**

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Immersive Digital Media – BSc

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Department of Information Technology

Technological University of the Shannon: Midlands Midwest



# TUS

**Technological University of the Shannon:  
Midlands Midwest**

Ollscoil Teicneolaíochta na Sionainne:  
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Date Submitted: 27/04/2025.

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# Table of Contents

|          |                                    |    |
|----------|------------------------------------|----|
| <b>1</b> | <b>Abstract</b>                    | 1  |
| <b>2</b> | <b>Acknowledgements</b>            | 2  |
| <b>3</b> | <b>Introduction</b>                | 3  |
| 3.1      | Project Overview                   | 3  |
| 3.2      | Problem Statement                  | 3  |
| 3.3      | Proposed Solution                  | 4  |
| 3.4      | Research                           | 4  |
| 3.4.1    | Target Audience                    | 4  |
| 3.4.2    | Users                              | 12 |
| 3.4.3    | Word Cloud                         | 14 |
| 3.4.4    | Competitors                        | 15 |
| 3.4.4.1  | Headspace XR                       | 15 |
| 3.4.4.2  | Tripp                              | 16 |
| 3.4.4.3  | Moment                             | 17 |
| 3.4.5    | Uses and Benefits of XR Technology | 17 |
| 3.4.6    | Drawbacks of XR Technology         | 18 |
| 3.5      | Technologies / Tools               | 19 |
| 3.5.1    | Figma                              | 19 |
| 3.5.2    | Firebase                           | 20 |
| 3.5.3    | VS Code                            | 21 |
| 3.5.4    | Convai                             | 21 |
| 3.5.5    | Blender                            | 22 |
| 3.5.6    | Spline                             | 22 |
| 3.5.7    | Bootstrap                          | 23 |
| 3.5.8    | ChatGPT-3.5 Turbo                  | 23 |
| 3.5.9    | GitHub                             | 24 |
| 3.6      | Programming Languages              | 24 |
| 3.7      | Hardware Requirements              | 24 |

|           |                                    |    |
|-----------|------------------------------------|----|
| <b>4</b>  | <b>Design</b>                      | 25 |
| 4.1       | User Interface Design              | 25 |
| 4.1.1     | Colour Theory                      | 25 |
| 4.1.1.1   | Colour Palette                     | 25 |
| 4.1.2     | Typography                         | 27 |
| 4.1.3     | Iconography                        | 28 |
| 4.1.4     | Moodboards                         | 29 |
| 4.1.5     | Logo Evolution                     | 30 |
| 4.1.6     | Sound Design / Audio Feedback      | 31 |
| 4.2       | Application Structure & Wireframes | 32 |
| 4.2.1     | Web App Architecture               | 32 |
| 4.2.2     | Web App Wireframes                 | 33 |
| 4.2.3     | Web App – Mid Fidelity Prototypes  | 38 |
| 4.2.4     | VR App Architecture                | 42 |
| 4.2.5     | VR Wireframes                      | 42 |
| 4.3       | Technical Design                   | 43 |
| 4.3.1     | Project Architecture               | 43 |
| 4.3.2     | Third-Party Integrations           | 44 |
| 4.3.2.1   | ChatGPT-3.5 Turbo                  | 44 |
| 4.3.2.2   | Convai / ReadyPlayerMe             | 44 |
| 4.3.3     | Backend and Database Design        | 44 |
| 4.3.4     | System Workflow                    | 45 |
| <b>5</b>  | <b>Implementation</b>              | 46 |
| 5.1       | Environment and Scene Design       | 46 |
| 5.2       | Interactivity and User Flow        | 52 |
| 5.3       | Voiceover and Audio Integration    | 53 |
| 5.4       | Technical Implementation           | 54 |
| 5.4.1     | Overview                           | 54 |
| 5.4.2     | Code Integration                   | 55 |
| 5.4.2.1   | Web Application                    | 55 |
| 5.4.2.1.1 | Firebase                           | 55 |

|           |                                                   |           |
|-----------|---------------------------------------------------|-----------|
| 5.4.2.1.2 | JavaScript Logic Examples.....                    | 61        |
| 5.4.2.2   | VR Application.....                               | 64        |
| 5.4.2.2.1 | Firebase .....                                    | 64        |
| 5.4.2.2.2 | Convai API .....                                  | 66        |
| <b>6</b>  | <b>User Testing .....</b>                         | <b>68</b> |
| 6.1       | Changes made based on User Feedback.....          | 68        |
| 6.1.1     | Web Application.....                              | 68        |
| 6.1.1.1   | Classmate Survey Tasks .....                      | 68        |
| 6.1.1.2   | Web App Changes .....                             | 68        |
| 6.1.1.3   | Think Aloud/Heuristic Evaluation – Feedback.....  | 69        |
| 6.1.1.3.1 | Positive Feedback Examples.....                   | 69        |
| 6.1.1.3.2 | Negative Feedback Examples .....                  | 70        |
| 6.1.2     | VR application .....                              | 70        |
| 6.1.2.1   | Friends - Positive Feedback.....                  | 71        |
| 6.2       | Screenshots – User feedback / Survey Results..... | 71        |
| 6.2.1     | VR App Feedback.....                              | 71        |
| 6.2.2     | Classmate Survey Results – Google Forms .....     | 71        |
| <b>7</b>  | <b>Challenges &amp; Reflection .....</b>          | <b>75</b> |
| 7.1       | Challenges.....                                   | 75        |
| 7.2       | Reflection .....                                  | 77        |
| <b>8</b>  | <b>Future Work .....</b>                          | <b>77</b> |
| 8.1       | Expanded Content and Zones.....                   | 77        |
| 8.2       | Personalisation and Progress Tracking.....        | 77        |
| 8.3       | Web Integration and Mobile Companion App .....    | 78        |
| 8.4       | Accessibility and Localisation.....               | 78        |
| <b>9</b>  | <b>Table of figures.....</b>                      | <b>79</b> |
| <b>10</b> | <b>References.....</b>                            | <b>82</b> |

# 1 Abstract

*Can immersive environments promote better digital habits in users?*

[Neuroloop](#) is an immersive VR experience that explores how social media influences brain chemistry and behaviour. Aimed at young adult users, the experience guides them through interactive zones that highlight key brain regions involved in attention, emotion, and reward – such as the Ventral Tegmental Area, Nucleus Accumbens, Prefrontal Cortex, and Amygdala. Users learn how dopamine functions, how neurons communicate, and how digital feedback loops can shape habits. The experience concludes with a short quiz that reinforces key concepts and encourages reflection. By combining neuroscience education with an engaging virtual environment, Neuroloop promotes a deeper understanding of the brain's role in online behaviour. A companion website extends the experience with additional resources, making Neuroloop a dynamic, multi-platform learning tool.

## **2 Acknowledgements**

I would like to express my sincere gratitude to everyone who supported me throughout my academic journey at the Technological University of the Shannon. Your patience, time, and expertise will stay with me throughout my career, and I can't thank you enough.

Special thanks to Lorraine Callanan, Elizabeth Bourke, Kelly O'Brien, John Jennings, and Seamus O'Ciardhiun. I wish you all the absolute best in the future.

I'd also like to extend my thanks to my classmates and the wider college staff – including the canteen team, reception, and everyone working behind the scenes for creating a welcoming and supportive environment that made day-to-day life on campus that much better.

## 3 Introduction

### 3.1 Project Overview

Neuroloop is a VR-based educational project that bridges neuroscience and interactive design to address the growing impact of social media on brain function and digital habits. Developed using Unity, Blender, and supporting technologies like Convai and Firebase, the experience places users in an environment where they can explore how social media platforms engage cognitive systems.

Through immersive scenes and audio-narrated content, users learn about dopamine, neural communication, and how brain regions like the Prefrontal Cortex play a role in attention and emotion online. I designed each zone to be informative yet approachable, finishing with a short quiz that reinforces learning outcomes. A companion website complements the VR experience, offering users insights and resources beyond the headset.

### 3.2 Problem Statement

The way people consume content daily – especially on social media – is increasingly shaped by algorithms designed to keep them engaged. While smartphones and platforms offer undeniable benefits, they also encourage compulsive use, reduced focus, and a reliance on external validation. For many users especially young adults, this has become an unconscious loop, reinforced by the dopamine-driven reward systems these platforms are built on.

My interest in immersive technology began when I first experienced the Meta Quest 2 in 2022. It sparked the idea that XR could be more than entertainment – it could become a powerful tool for education and behaviour change. For this reason, I asked the college if it would be possible to order a newer version of the headset I used in 2022, the Meta Quest 3, which has more capabilities than its predecessor. Thank you to TUS, and to Lorraine Callanan in particular for making this idea become a reality.

In this project I wanted to address a growing issue that people, especially students, often lack awareness of how social media affects their brain, decision making, and well-being.

Studies show that high levels of media multitasking are associated with reduced grey matter in the anterior cingulate cortex and other prefrontal area – regions responsible for attention, impulse control, and working memory ([Firth et al., 2020](#)). Despite this, most digital well-being education fails to fully engage users or explain the science in a meaningful way.

There is a pressing need for an immersive and accessible experience that helps users visualise what's happening in their brain when they engage on social media – and reflect on how these platforms influence their attention, emotions, and habits. Neuroloop is my attempt to meet that need.

### 3.3 Proposed Solution

An immersive virtual reality experience designed to educate users on the neurological impact of social media in an engaging and accessible way. By combining interactive 3D environments with narrated scientific explanations, the experience aims to help users explore the brain's key regions involved in reward, attention, and emotion.

Through visual metaphors, animation, and voice overs, users are guided through how digital platforms are designed to keep them hooked. To conclude, the experience will finish with a short interactive quiz that reinforces learning and self-reflection.

A companion website with additional resources and support, allowing users to revisit key concepts outside of the headset will help with further reinforcement. By merging education with immersion, the project aims to improve digital awareness and promote healthier relationships with technology.

### 3.4 Research

#### 3.4.1 Target Audience

While I'm sure we can all agree that social media overuse is a growing problem for people of all ages, for this project, I've chosen to primarily aim at third-level students, particularly those who may not have studied science beyond third year in secondary school. This demographic is highly active on social media platforms and often unaware of the neurological mechanisms influencing their online behaviour.

The project is designed to present neuroscience concepts in a visually engaging and easy-to-understand format, making it accessible to non-scientific audiences. By using immersive technology, Neuroloop appeals to student's curiosity and familiarity with digital experiences, while promoting critical reflection on their personal tech habits.

The tone, visuals, and language throughout the experience are tailored to be informative without being overly academic, ensuring that users can connect with the content regardless of their educational background.

To gain insight into my audience, I conducted an anonymous ten question survey using Google Forms in order to collect data. It seemed only relevant to distribute the survey through social media channels seeing as it's the fastest way to reach a broader audience. For this reason, I made social media posts on Reddit and focused on specific subreddits (pages more likely to be used by my target audience) – r\College, r\University, r\VrDev, r\Unity. I also used my personal Instagram account to share the survey and sent it to various WhatsApp groups and friends and family. I did my best to collect as much data as possible and ended up with 26 responses for my survey. Below are my findings.

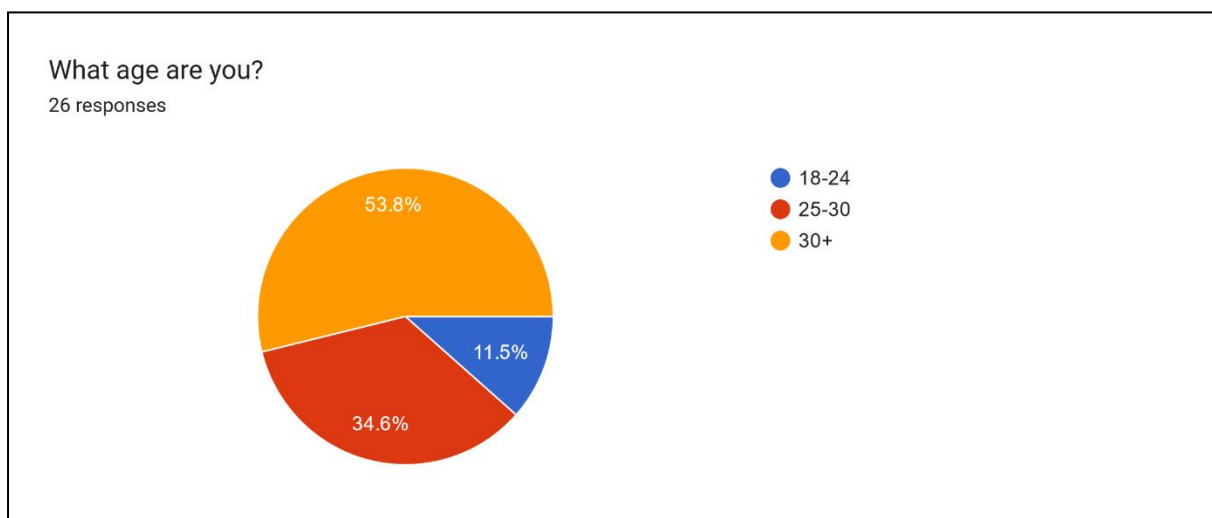


Figure 1 – Survey q1.

Given that I'm a mature student, it's hardly a shock that the majority of my responses from figure 1 were people over the age of 30 as I posted through my own social media channels, from this I realise there are limitations to my survey. In total, I received 14 or 53.8% responses from people 30+, 9 or 34.6% responses from participants aged 25-30, and 3 responses from participants aged 18-24.

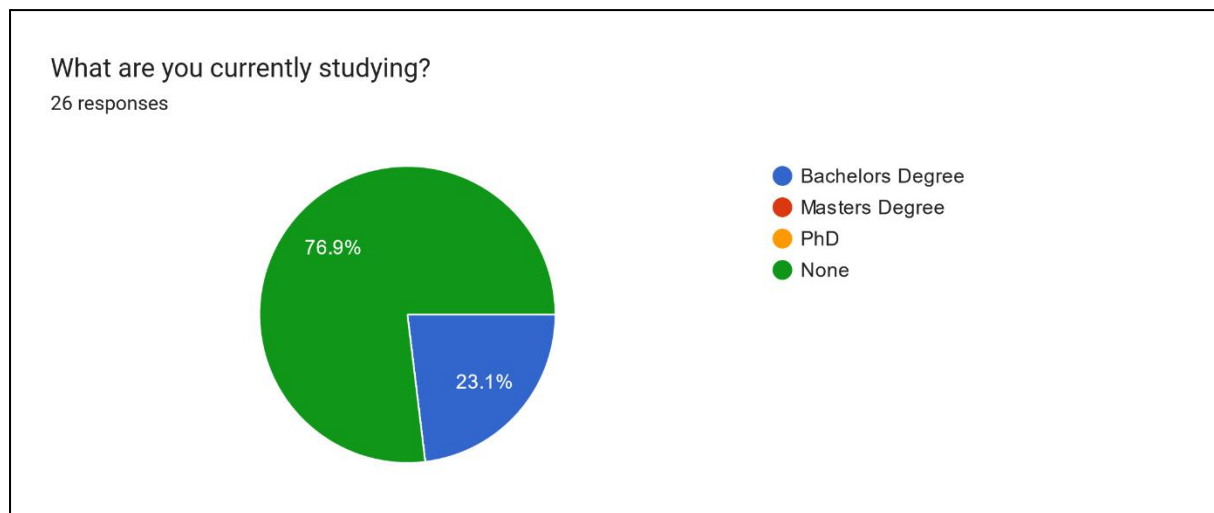


Figure 2 – Survey q2.

Unfortunately, I didn't get as many third level students as I had hoped to complete the survey. This had led me to a crossroads and wondering if I should change my target audience altogether but seeing as I'd like to test my prototype on students, I will persevere with what I have got. 23.1% or 6 bachelor students took part in the survey, while the majority of 76.9% or 20 other participants were not students at all.

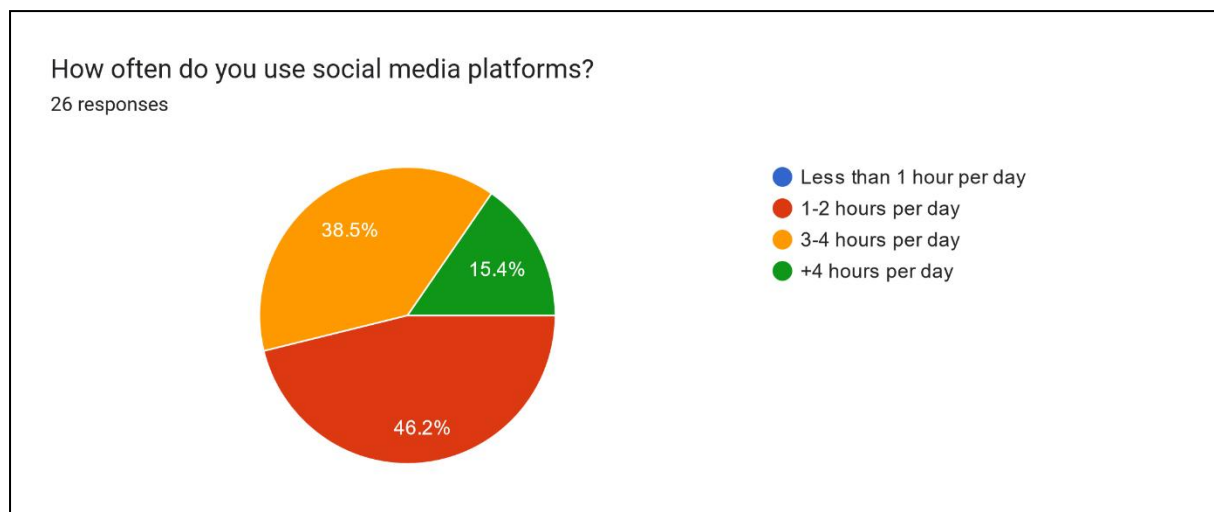


Figure 3 – Survey q3.

It was interesting to see that out of 26 participants, none of them use social media platforms for less than an hour. Leaving me to believe that if we all were really tracking our usage on a weekly or monthly basis that we could change a lot of that usage into better spent time. 38.5% or 10 participants use it for 3-4 hours/day, 46.2% or 12 participants use it for 1-2 hours/day and 4 participants use it for 4+ hours/day.

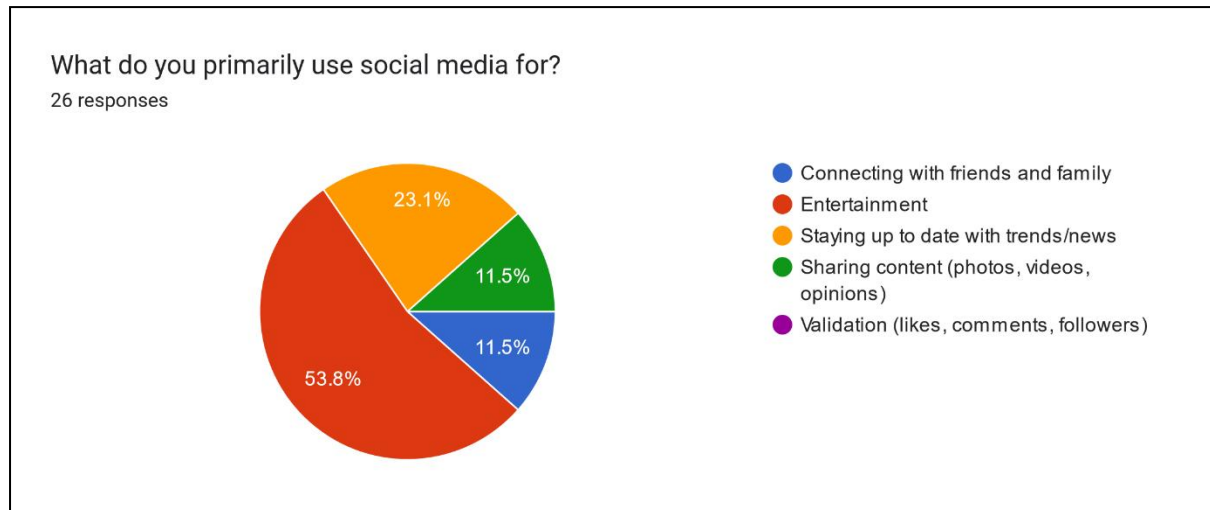
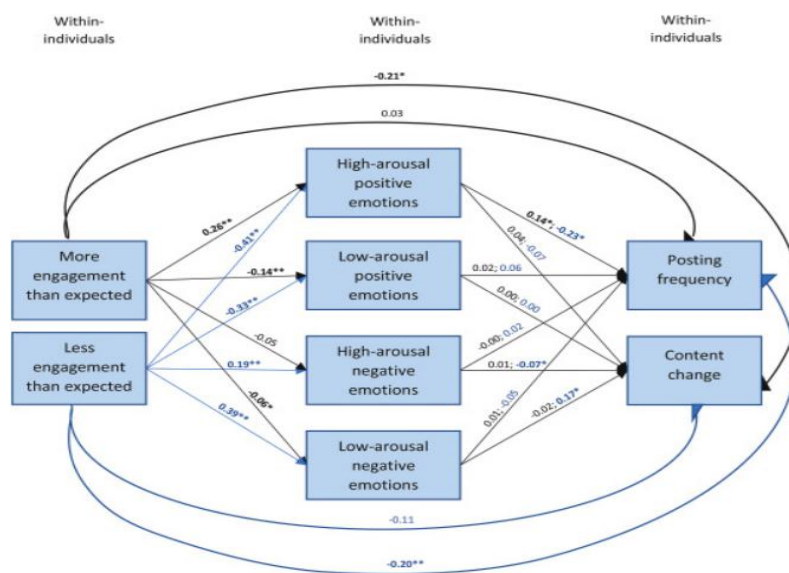


Figure 4 – Survey q4.

This survey question was aimed at understanding why people use social media. Surprisingly, none of the 26 participants described using social media primarily for validation purposes. However, research by (Stsiampkouskaya, Joinson and Piwek, 2021) suggests a distinct perspective, indicating that ‘high-arousal emotions, such as excitement, can drive frequent posting, suggesting that these positive emotional spikes reinforce social media engagement.’ This is relevant to my project because it aims to educate the users on validation-driven behaviours. My goal is to mitigate user habits that are rooted in emotional reinforcement. By developing XR simulations to bridge this gap in user knowledge this project has the potential to cause positive behavioural change.



\* Significant using a 95% credible interval

\*\* Significant using a 99% credible interval

Figure 5 – Research (Stsiampkouskaya, Joinson and Piwek, 2021).

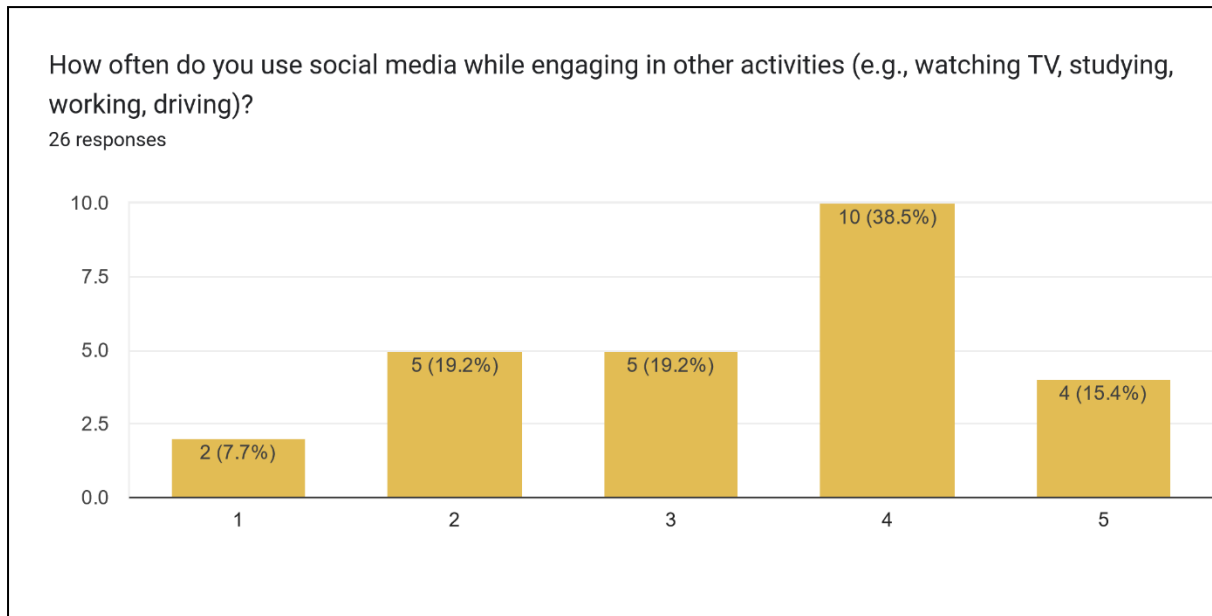


Figure 6 - Survey q5.

The participants answered from 1 – 5, one being never and five being always. The results point to the fact that people are more likely to be using their phones while engaging in other activities.

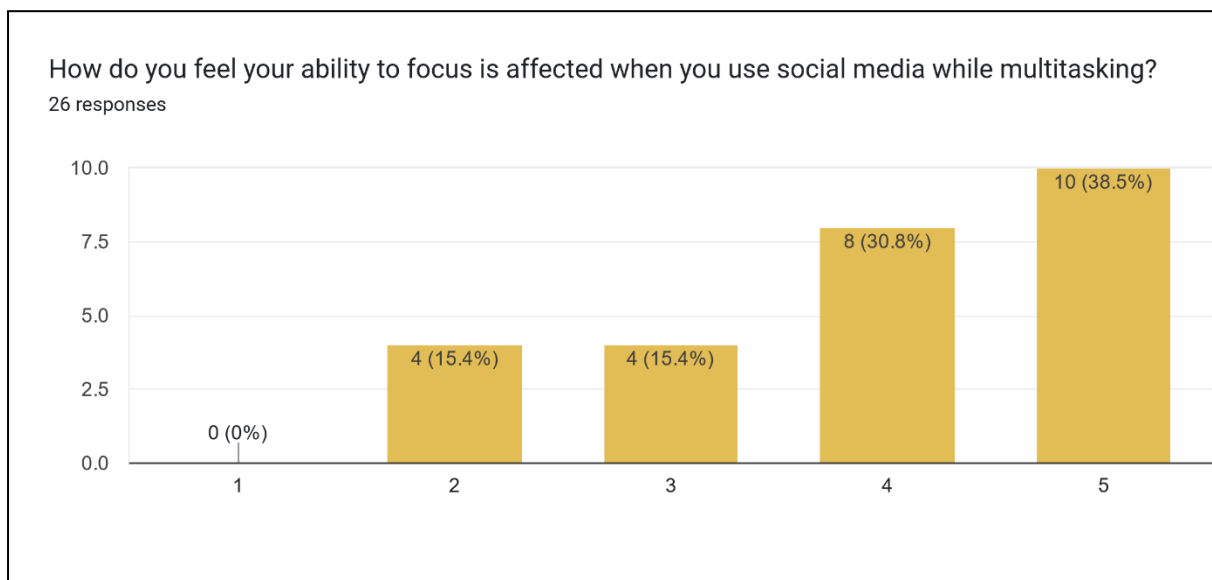


Figure 7 - Survey q6.

Of all the results so far, this was undoubtedly the clearest cut. Participants rated 1-5, one being never and five being always. It demonstrates that although the majority of people understand that their focus is being affected by using social media while

watching television for example, they continue to do it and leads me to believe a habit could be causing this behaviour. Not one person said that their focus wasn't affected highlighting a need to raise awareness and promote behavioural change. In contrast to this evidence, the paper by (Baumgartner, 2022) outlines the fact that *'the key question of whether digital media causally impairs attention cannot yet be conclusively answered. To answer this question, it is crucial for the field to advance the theoretical as well as methodological approaches that we currently employ.'*

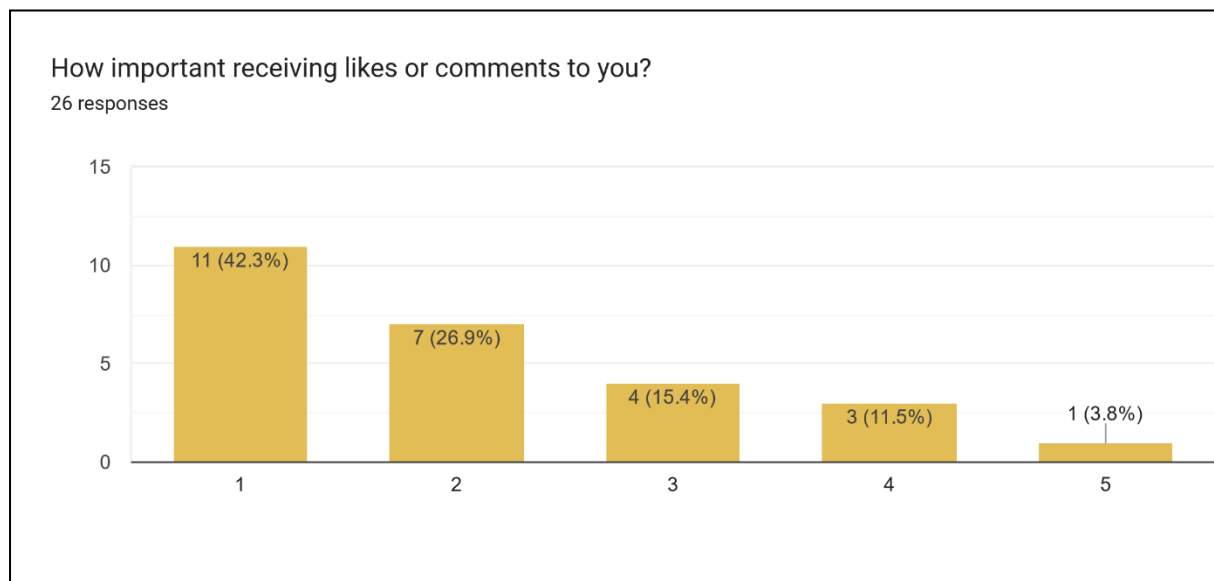


Figure 8 - Survey q7.

It was pleasant to see a downward trend indicating that people were less inclined to give importance to receiving likes or comments. Regardless of the results of the 26 and to back up my project goal, I still think that validation-seeking behaviour is still worth addressing. In my opinion the results of this question do not reflect true reality. Whether people wish to believe it or not evidence shows that *'The implications of social validation for individuals are indeed far-reaching. Users often seek social validation on social media platforms to enhance their self-esteem, establish their social status, and reinforce their sense of belonging within their online communities. Likes, comments, and shares serve as tangible indicators of acceptance and popularity, providing individuals with a sense of validation and recognition.'* (Ballara, 2023).

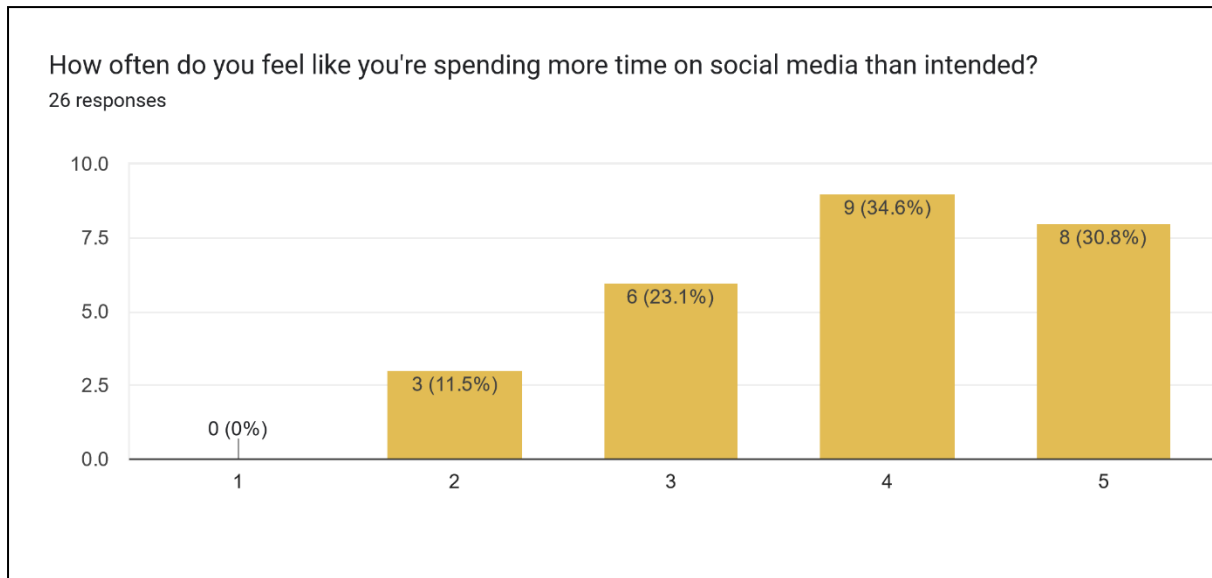


Figure 9 - Survey q8.

This chart shows that the majority of respondents are spending more time on social media than they intend to, highlighting a need for better self-regulation.

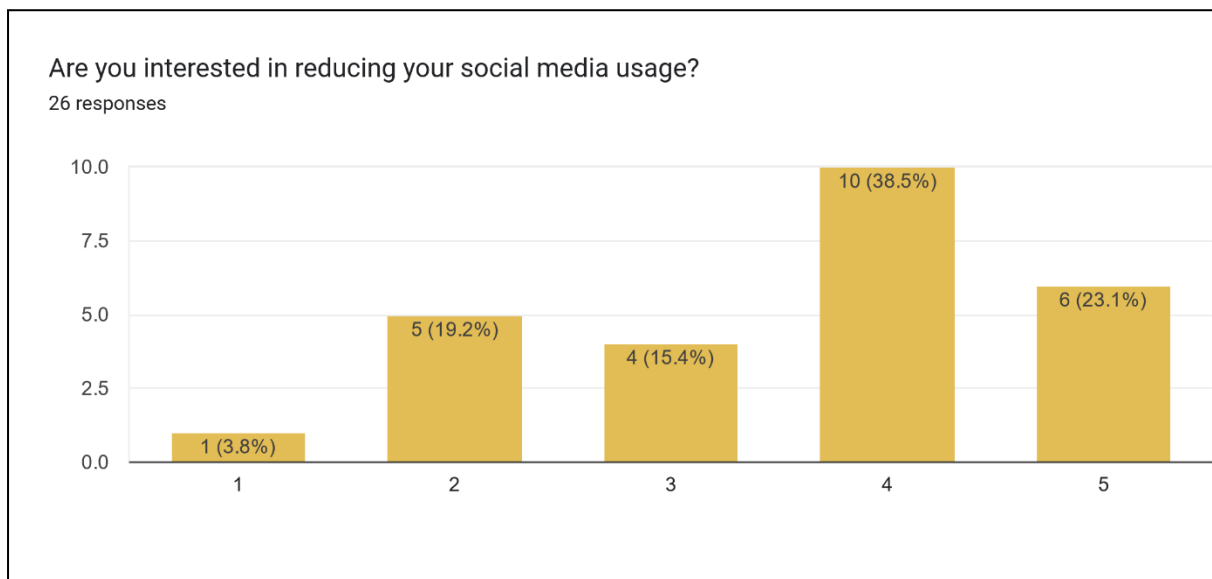


Figure 10 - Survey q9.

Looking at these results, it has to be said that it's good to see that some participants are happy with how they manage their social media usage. However, again, the figures point to a majority of people who are interested in reducing how much they use social media, which in turn points to the gap in the market for a tool promoting more controlled social media use.

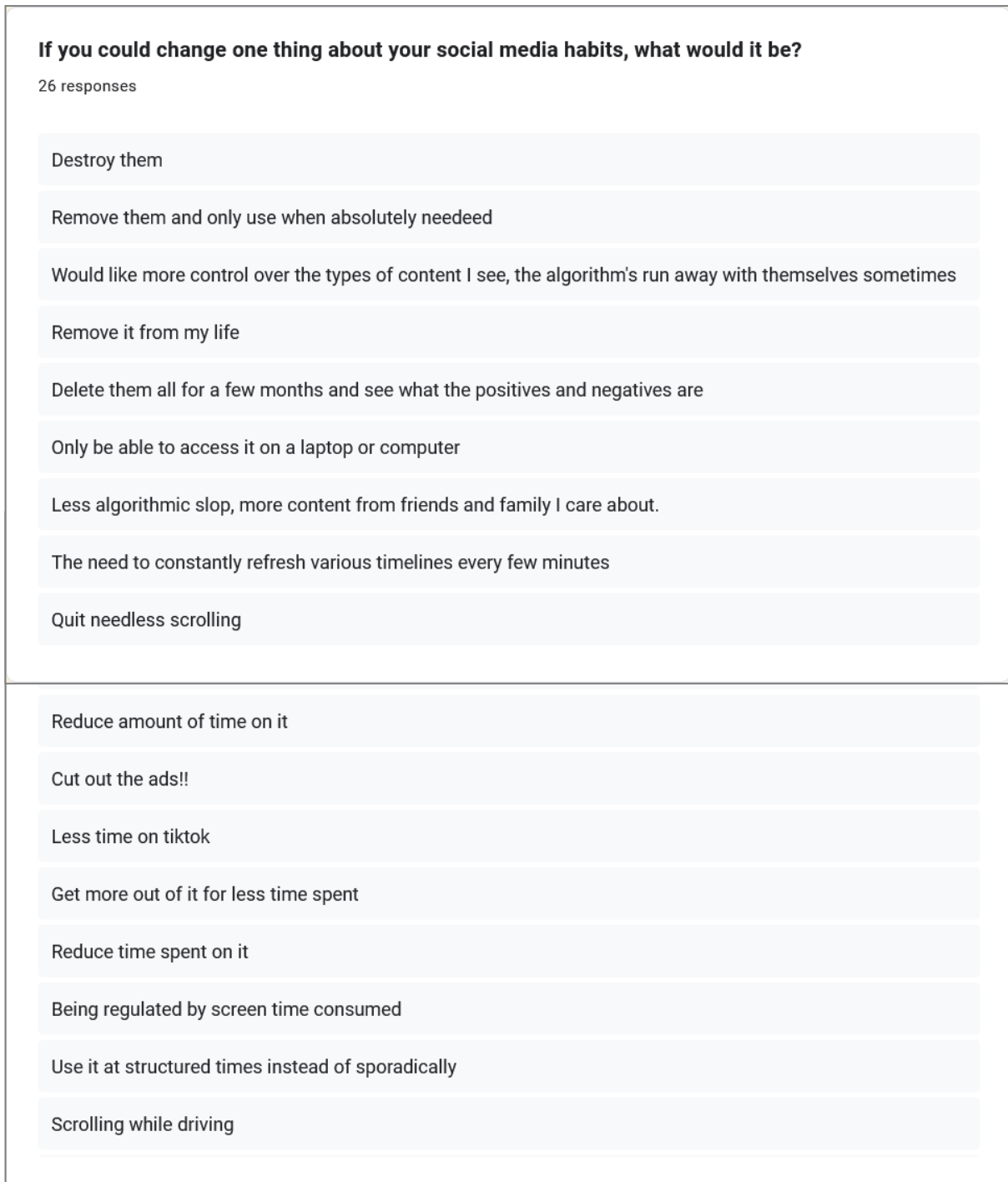


Figure 11 – Survey q10

### 3.4.2 Users

To get a better picture of who may use the application I created a user persona. This would ideally help me to gain an understanding of my target audience and build empathy to focus on their experiences/challenges when using the application, rather than making my own assumptions. This helped me to define the requirements of my website and application and to avoid scope creep by filtering out unnecessary features that don't align with my user needs.



Figure 12 – Persona 1

The second persona that I designed is Jack Smith, who is a young engineering student. His core needs are to manage his time more effectively, to relax, and to feel financially stable. Jack's main frustrations are managing time between activities, his finances for new gaming gear, and his focus on college work which is disrupted by various mobile notifications.

### Jack Smith



|               |                    |
|---------------|--------------------|
| AGE           | 20                 |
| EDUCATION     | BSc in Engineering |
| STATUS        | In a relationship  |
| OCCUPATION    | Student            |
| LOCATION      | Dublin             |
| TECH LITERATE | High               |

#### Personality

Introverted

Gamer

Saver

Tech-savy

#### Bio

Jack currently lives in Dublin with his family. He enjoys playing video games and often plays online with his friends. On the weekend Jack likes to relax by catching up on tv shows with his girlfriend. He is also a member of his local hurling team and is a part of many social media groups.

#### Core needs

- To manage time better
- A outlet to relaxation
- Feeling financially stable

#### Frustrations

- Struggling to focus on college work due to notifications
- Managing time between different activities
- Limited budget to get gaming gear

#### Brands



Figure 13 – Persona 2

### 3.4.3 Word Cloud

In order to visualise the data collected from the open ended question on my Google Forms survey I conducted "What would you change about your social media habits?," I copy and pasted the text to a [Free Word Cloud Generator](#) and the following image was the result.

The word cloud represents the data as larger if the word is more frequent and smaller if less frequent. Although I had a relatively low number of responses at 26, it's easy to see that the resounding change needed regarding social media habits it's, less, time, and scrolling.

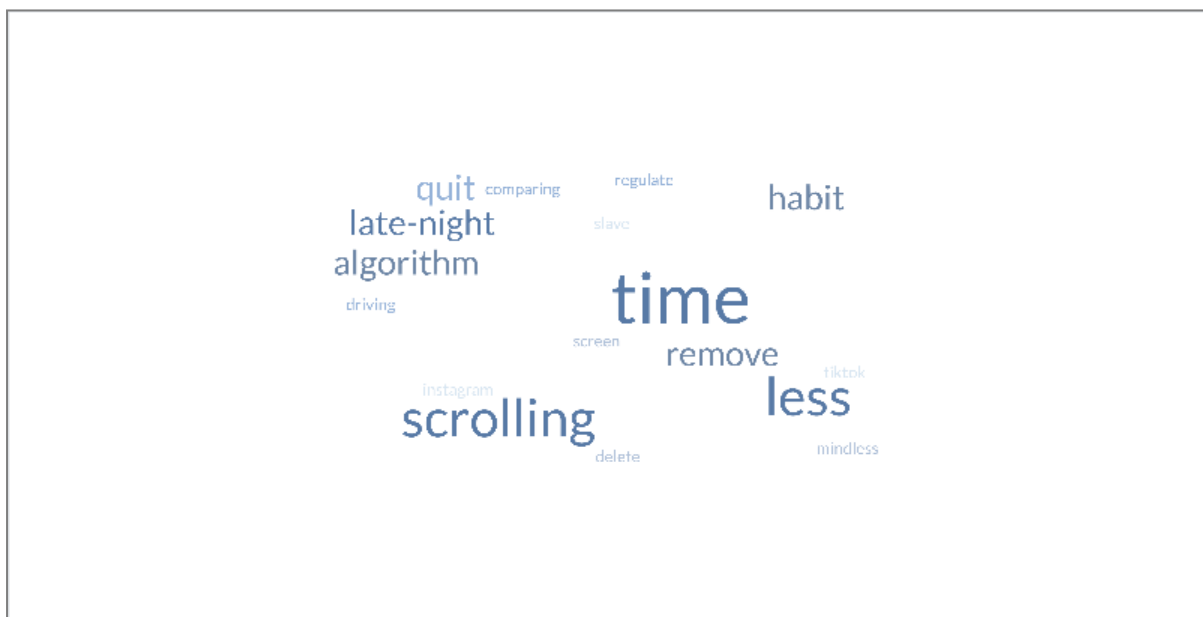


Figure 14 – Word Cloud

### 3.4.4 Competitors

To get a more complete picture of the application I wanted to make, I researched into a number of would-be competitors in the XR space whose aim is to tackle their user's habits. Seeing as the Extended Reality market is quite new, I found it difficult to find companies in direct competition with what I'm trying to achieve. The first company I found that related to my project was headspace.

#### 3.4.4.1 Headspace XR

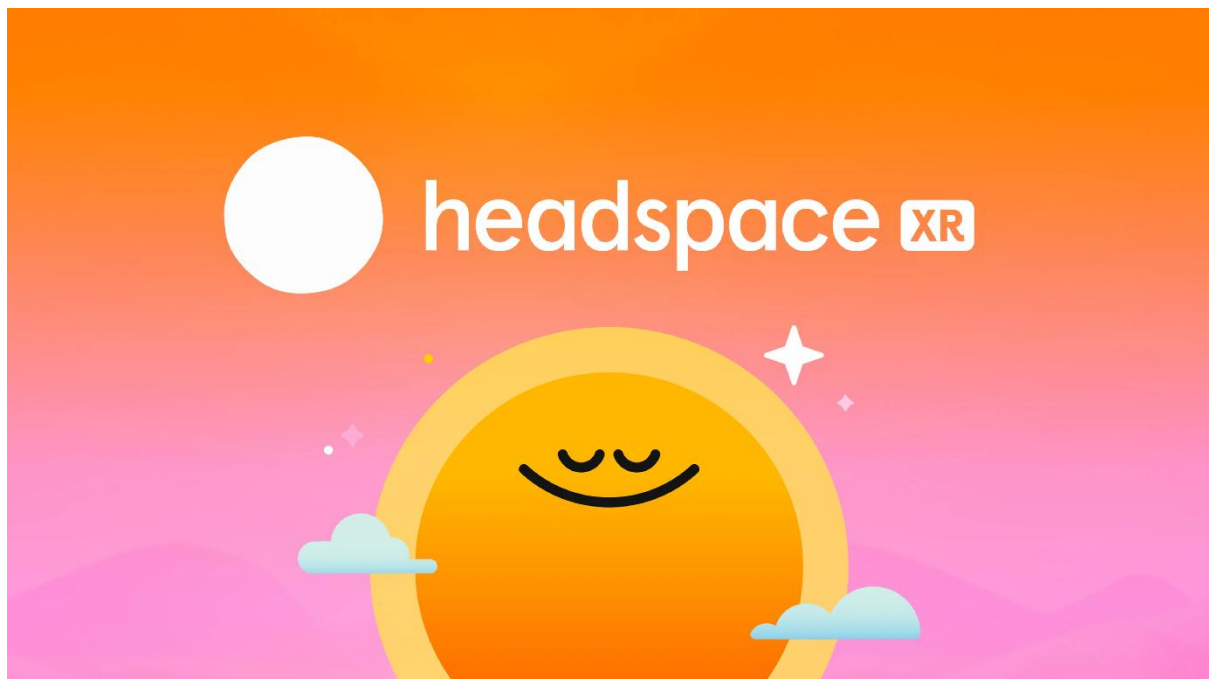


Figure 15 – headspace.

[Headspace](#) is a mindfulness application that help users by teaching them different techniques. Their goal is to create an environment to equip their users with the skills necessary to promote their own mental wellbeing and with that regard I would like to do the same with my app. Their unique selling point is making mental health support accessible to all – I don't have the flexibility as XR headsets are not commonplace in every household (yet) but the Web app I create will be accessible to all. By targeting areas like sleep, breathing, meditation, and movement, their app helps the user create life-changing habits, again it aligns with my concept. The application started primarily on mobile but has later transformed into other media sources such as Netflix, and more recently to the XR world. Having used the application myself, I think it's really well done and great asset to the world today when modern life is so busy and compact. In my opinion, headspace definitely helps

one learn to switch off. They have partnered with Meta to bring their mental health support to the virtual world. I would like to draw some inspiration from this app and help my users turn skills learned in the virtual world into skills that can be applied in the real world.

#### 3.4.4.2 Tripp



Figure 16 - Tripp.

[Tripp](#) is a virtual reality application used to promote mindfulness and wellbeing for their users. It's quite similar to Headspace in a sense but there are nuances, and they do take a unique style and approach to their counterpart. Tripp is available to download to your mobile device from the Google Play or Apple App Store. They are also compatible with a number of different devices like the Pico, Vive Flow, Meta Quest 3, and Apple Vision Pro. They have an interesting AI-powered wellness coach called 'Kōkua' that is trained by the Tripp community and can give real-time feedback if you tell it what is on your mind. Some of their other products include, activity tracking to check on your habits and moods, guided meditations, sleep stories, and Escapes which place you in a virtual greenspace with nature sounds, which has been proven to better one's mood (Song et al., 2023). Along with these other features, Tripp facilitates virtual meditation events to allow people to feel as part of the community and less isolated.

### 3.4.4.3 Moment

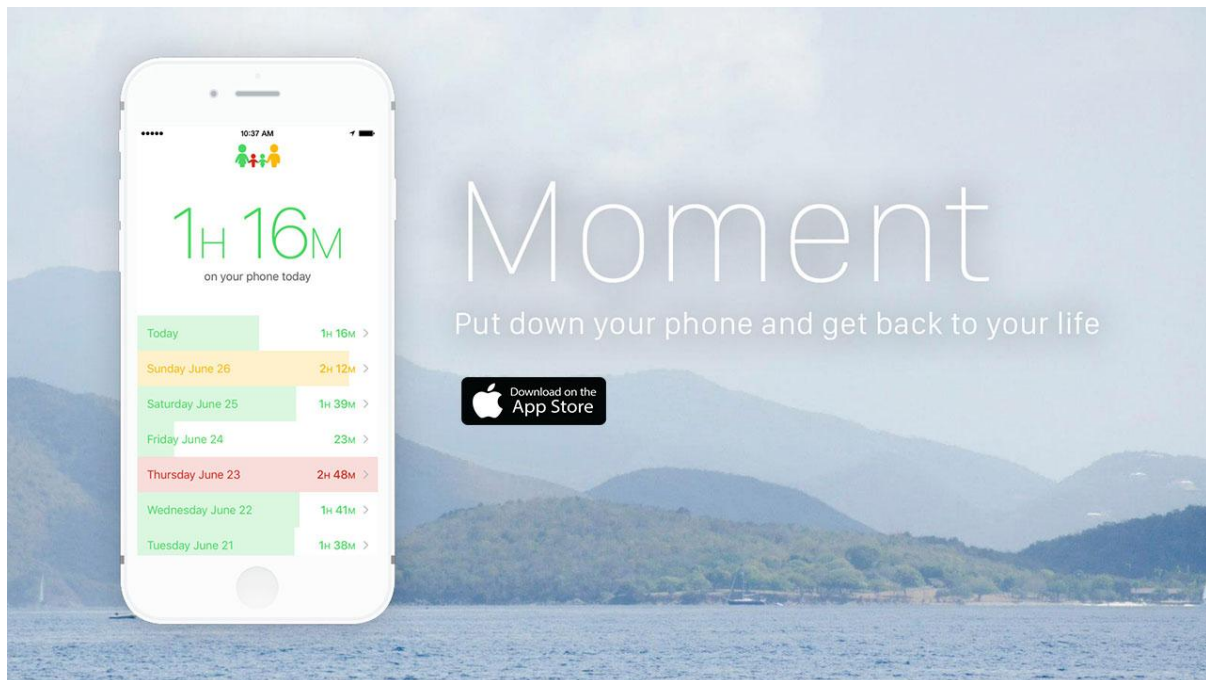


Figure 17 - Moment.

[Moment](#) is a mobile application available on the iOS App Store. As you can gather from the image above Figure 15, it's an app with one goal in mind, which is helping people to limit their usage and screen time on their mobile phones and get back to living 'normal' life without screens. This app has an interesting feature that allows you to manage your family's screens. It lets each participant to view the screen usage of all other participants, it also gives them the ability to set a timer and lock screen usage for example when watching a movie or eating dinner together. They hope to improve wellness and focus by limiting screen time each day. Although this is not an XR application, it fits in with my goal of making people think critically about their own screen time and habits.

### 3.4.5 Uses and Benefits of XR Technology

XR technology has many advantages across various industries. It can be used to train staff for jobs which are dangerous (e.g., working on oil rigs, firepersons, electricians) without a fear making a fatal or expensive mistake. Using this technology in the classroom can promote learning through interactive engagement and give students a unique perspective on what it is they are learning and improve lesson comprehension and retention. The following heading and chart are results taken from (Zolezzi *et al.*, 2024). It shows how 26 students rated the effectiveness of XR technology at helping them understand their lessons.

*“The XR technology used in the 360-degree image-based units helped me to better understand the content of the lessons”.*

Ho avvertito che i contenuti delle lezioni nelle unità basate su immagini a 360° erano ben strutturati e facili da seguire.

26 risposte

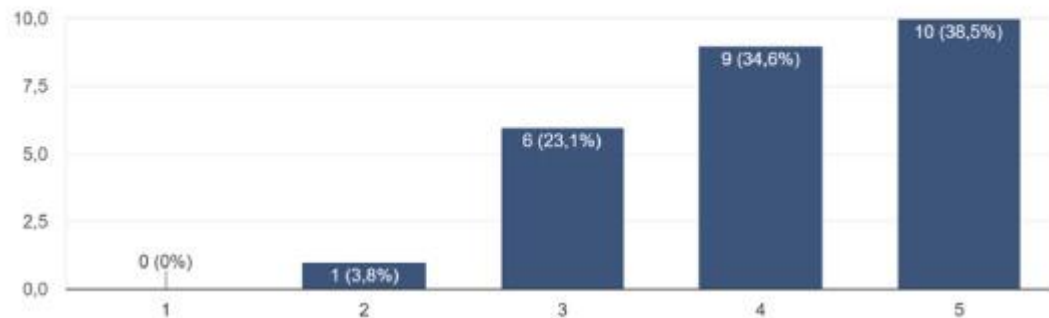


Figure 18 – Research (Zolezzi et al., 2024).

XR technology can allow medical students and professionals to practice procedures and understand complex anatomy without the need for test subjects. Or, be used to simulate battle scenarios to safely improve military defences. XR technology makes visualising and scaling products, buildings or designs easy and can save costs, time, and effort.

### 3.4.6 Drawbacks of XR Technology

Like everything, there is always a downside. The biggest hurdle for most small organisations is the costs, the hardware and software needed is often too expensive to allow them to implement it. There is usually a limit on the environment, some Head Mounted Displays need specific lighting requirements or room setups. There is often a limited battery life and comfort after wearing the headsets for extended periods can be questioned. Prolonged use can lead to potential eye strain and/or motion sickness. On a personal level, I can account for the promotion of eye strain.

## 3.5 Technologies / Tools

This is a list of the technology stack and tools I used when creating both the VR application and the companion website.

### 3.5.1 Figma



*Figure 19 – Figma logo.*

[Figma](#) is a cloud-based interface design and prototyping tool that enables real-time collaboration among team members. It's designed for UI/UX designers, developers, and product teams. It allows users to create, test, and iterate on interactive designs, and the best part is you can use it directly in the browser. Everything you work on is saved automatically, and it also has a mobile app that allows you to view your work (when you're not designing on the laptop) which gives you greater accessibility and a different perspective on whatever project you have been working on. This tool was instrumental for my design and development process. It's features and access to third-party tools helped me to constantly iterate and plan what I was doing next and also gave me access to wider resources quickly. I used it mainly for designing, planning, exporting assets that I produced, and editing assets that I had found that didn't quite fit the need for my project.



*Figure 20 – Unity logo.*

[Unity](#) is one of the leading games engines in today's market. Having used it before, it was a no brainer when it came to deciding on a technology that could be used to make a VR application for my final year project. To prepare and learn more about it, I participated in the annual Unity – Unite conference, which was held in September in Barcelona, 2024. This allowed me to meet with professionals throughout the gaming, engineering, and automotive industry and to attend a number of interesting talks and live demonstrations, revealing the newest updates (Unity 6) and what direction the company hopes to go in the future. It was an eye-opening experience, and I got to network and see how Unity is being used in many other ways, not just for creating games. This opportunity led me to focus on using Unity to create an application that can help the world without necessarily being a game.

### 3.5.2 Firebase



*Figure 21 – Firebase logo.*

[Firebase](#) is a comprehensive application development platform provided by Google that offers a suite of tools to help developers build, improve, and grow their apps. It supports development by integrating functions like firestore, cloud storage, cloud functions, user authentication, hosting, and analytics. This was a huge part of my project, it helped to sew everything together, and gave it the dynamic aspect that I desired, I'll explain further in the Technical Design section below.

### 3.5.3 VS Code



Figure 22 – VS Code logo.

To accompany Unity, I used [VS Code](#). This is an Integrated Development Environment (IDE) built by Microsoft. It allows you to build, edit, and debug code. Although I have little (if any) experience using C sharp (coding language) it's similarities to Java (which I have studied) allowed me to get by and I was able to achieve my intended goals for my project. I also used VS Code as the IDE to develop my website, I used HTML, CSS, JS, and Bootstrap. I used the CLI (Command-Line-Interface) to deploy my code to Firebase.

### 3.5.4 Convai

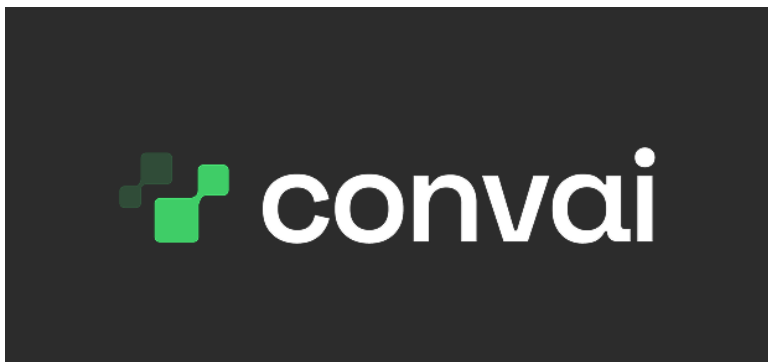


Figure 23 – Convai logo

[Convai](#) is an asset I found on the Unity Asset Store. I wanted to create a character for the VR app that would interact with the user, or act as an outlet if they needed guidance. This Unity plugin allows you to create an character via there affiliate companies website [Ready Player Me](#), which you can then import into your Unity project. Although this took a bit of tinkering, I was pleased with the end result. On the Convai website you can customize how you want your character to be, you choose a voice, a large language model, and you can set the limit of your character's knowledge by putting the information of you want them to know into their

'Knowledge Bank'. You can set up trigger for actions and animations, and in the end, you have an AI character that your user can talk to in game and get responses. I called mine Nancy.

### 3.5.5 Blender



Figure 24 – Blender logo

For a lot of my project I relied on premade assets from the Unity asset store, or from platforms like [Sketchfab.com](https://sketchfab.com), but the reliance only got me so far. Since I chose to use the Meta Quest 3 and to produce a standalone App, I needed to keep the size of my scenes within reason. When I chose a lot of assets they looked fine when I imported them into my scene, but were too large at run-time, and caused the app to stutter and crash. In these instances, I turned to [blender](https://www.blender.org/) to create my own assets and imported them into my Unity project. Additionally, to create videos (hooks) for my app, I recorded in the Meta HMD and imported the video into blender to edit and render the videos. I did the same for the mobile phone interactable video in the VR app 'Learn about the brain' scene.

### 3.5.6 Spline



Figure 25 – Spline logo

With this project I wanted to cause change. To do that, I needed something that would grab the user's attention from the get-go and appeal to their curiosity. [Spline](https://spline.design/) is a 3D software that does just that. It works in the browser or you can download it to your computer. Using it allows you to design 3D scenes and interfaces, add interactions (like hover effects, clicks, animations), and to export or embed them for

the web. This helped me to import the creepy girl that can't stop looking at her phone on my desktop version of my website - [neuroloop](#).

### 3.5.7 Bootstrap



*Figure 26 – Bootstrap logo*

[Bootstrap](#) is a free, open-source front-end framework for building responsive and mobile-first websites. It provides a collection of CSS stylesheets, Java Script components, and prebuilt UI elements. In the beginning of my website development, I was really struggling with Media Queries and making the site responsive, Bootstrap and it kicked my development into another gear. I used AOS (Animate-On-Scroll) for the animations on the homepage. I edited and overrode the Bootstrap library CSS, as necessary.

### 3.5.8 ChatGPT-3.5 Turbo



*Figure 27 – ChatGPT logo*

The project called for the use of an external API. Technically, the Convai plugin is using an external API, but I wanted to add a way that users could ask questions related to the topics covered if they were signed into their website profile. For this I purchased a [ChatGPT](#) subscription, created and saved the OpenAI API key, installed the necessary Google cloud functions and Firebase tools in my VS Code project CLI (and aforementioned consoles) and displayed the chat in the user profile using HTML, Java Script, Bootstrap, and CSS.

### 3.5.9 GitHub



Figure 28 – GitHub logo

[GitHub](#) is a platform for developers to store, share, and collaborate on code. I made a GitHub repository to create a backup of my Web/VR App files and allow me to show the project to future employers.

## 3.6 Programming Languages

1. HTML5
2. CSS/Bootstrap
3. Java Script
4. C#

## 3.7 Hardware Requirements

1. A laptop with a powerful graphics card, necessary RAM.
2. Meta Quest 3 HMD.
3. Link cable for testing access.



Figure 29 – Meta Quest 3

## 4 Design

### 4.1 User Interface Design

#### 4.1.1 Colour Theory

To influence my viewers' perception, I made use of colour theory. The idea being to use warm colours to induce happier and welcoming emotions. The applications concept is based around engagement and education. By incorporating colours of this nature I plan to capture attention and promote a positive learning atmosphere that encourages interaction and retention of information. Research from (Xia *et al.*, 2023) suggests how colour can be used as a design tool to enhance cognitive performance and emotional engagement in immersive virtual reality environments. It shows how specific colour attributes like hue, chroma, and brightness, can improve users' responses. A key finding was that the brightness of green can significantly increase the level of arousal and impulsiveness. At 75% brightness, participants made fewer errors and responded faster than those exposed to lower levels of brightness. These are all factors I took into consideration.

##### 4.1.1.1 Colour Palette

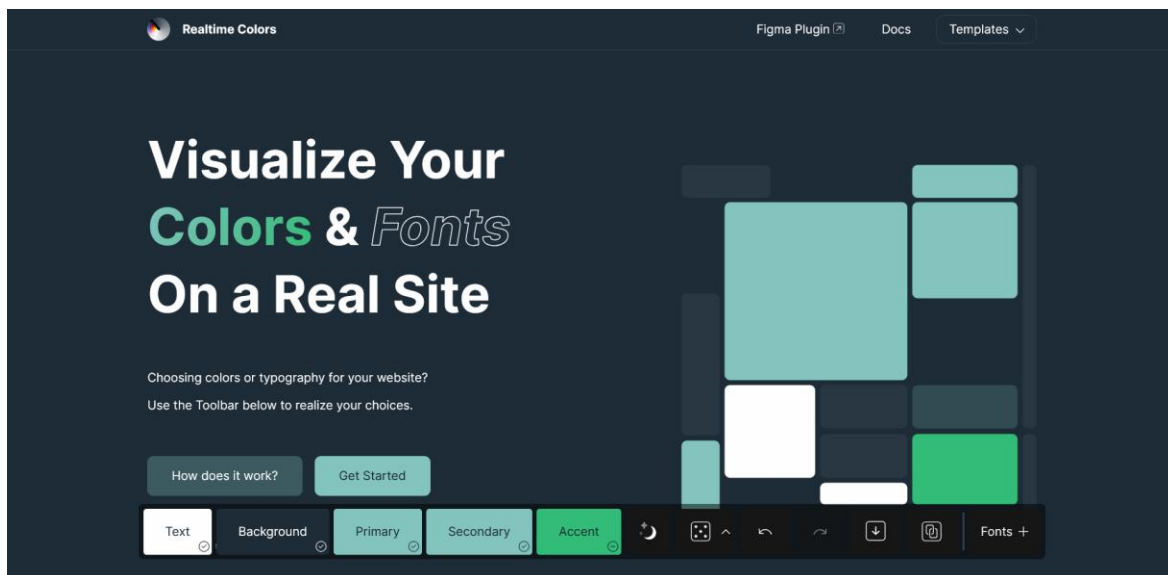


Figure 30 – Realtime Colors

Since I wanted to keep my Web App and VR App uniform, I had to take into account that extended time spent using HMDs can cause eye strain.

When researching online I found websites that had free colour palettes and spent time filtering through ones that would be appropriate. To speed up my workflow I experimented with colour palettes out on [realtimecolors](https://realtimecolors.com/) to give me a quick indication if it was the look I was going for and made mock-ups in Figma. From my research I knew that I wanted to have a dark background (without being too dark and causing a major contrast) and complementary soft greenish colours to promote a calm learning environment. The result in the figure above. A dark blue for the backgrounds, a soft teal as my primary colour, an off white for the text, and soft yet sharp green to give highlights and areas of interest pop. I also made use of the contrast plugin on Figma to see if the colours would pass the accessibility tests, see below.

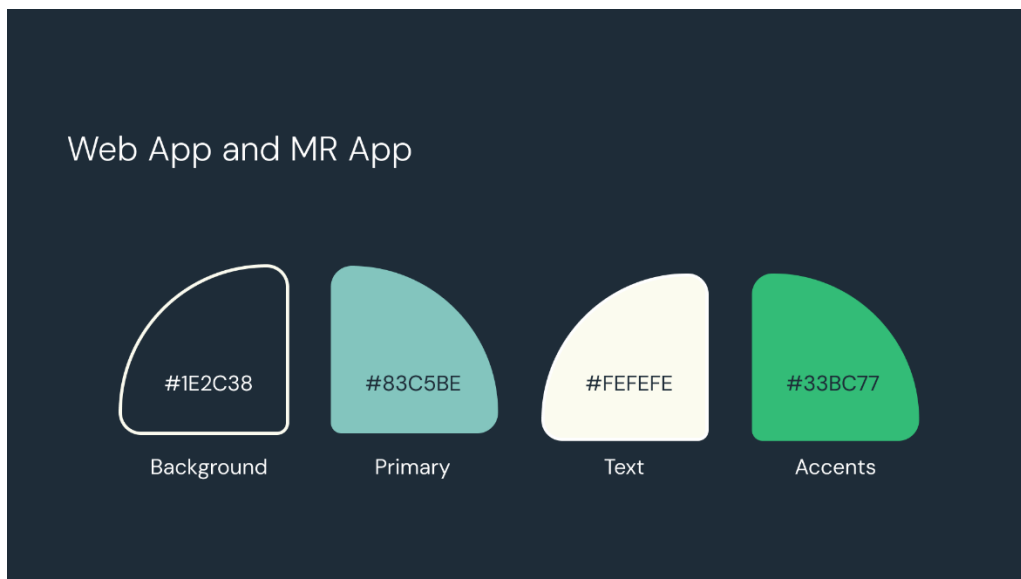


Figure 31 – Chosen Colour Palette

Below is an example of how I used Figma to ensure accessibility for my users.

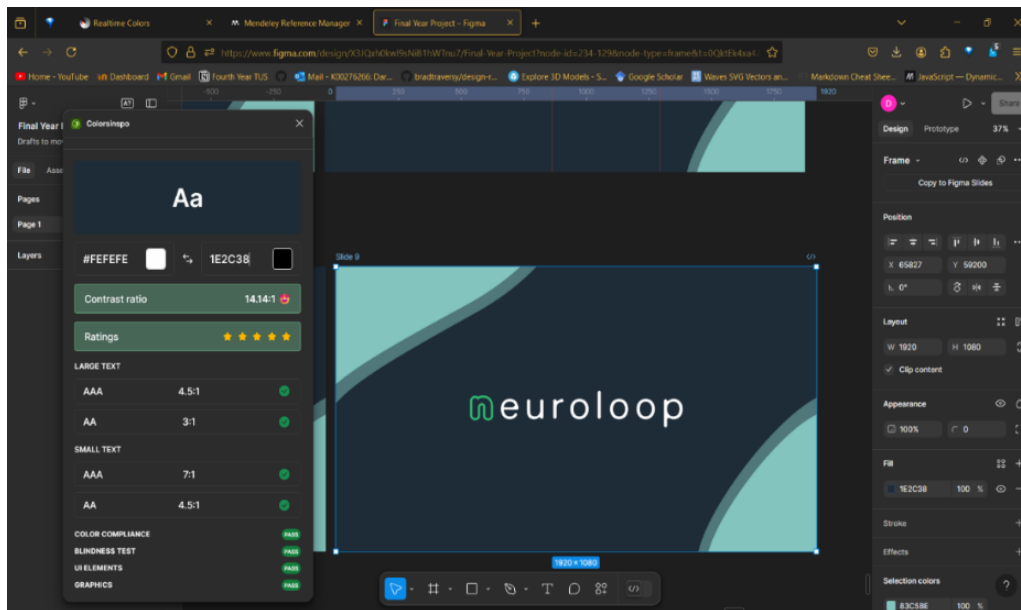


Figure 32 – Accessibility Checking

#### 4.1.2 Typography

The goal of my application is to educate the user. The priority of my chosen typeface was to ensure that all text was as legible, clear, and visually appealing to the viewer as possible. For all of these reasons, I chose DM sans which is a low-contrast, geometric sans-serif font designed for clear readability. It brings a simple, modern, and professional aesthetic that aligns well with my theme of neuroscience and education. It also adheres to typographic standards that enhance accessibility and is part of Google Fonts, meaning that it is open-source, free, and easy to integrate into my web and Unity applications. With simplicity in mind, I decided to use DM Sans on its own (without a pairing) and to use font style, size, and colour, to provide hierarchy, contrast, and clarity.

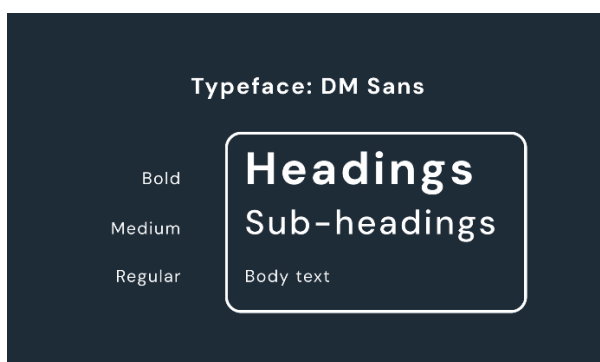


Figure 33 – Typeface – DM Sans

### 4.1.3 Iconography

To ensure consistency across the VR application and the Web app, I aimed to maintain a unified approach to iconography wherever possible. For both applications, I used a free icon library called [SVGRepo](#). Initially, I had planned to introduce 3D icons and buttons for the VR app, but during development I transitioned to using flat UI instead. The 3D buttons proved unnecessary and introduced challenges with the input system for the Meta controllers. The switch to flat UI icons simplified the interface and improved usability without compromising the immersive experience. I chose universally known icons to increase the ease of navigation and support users' understanding of interaction outcomes.

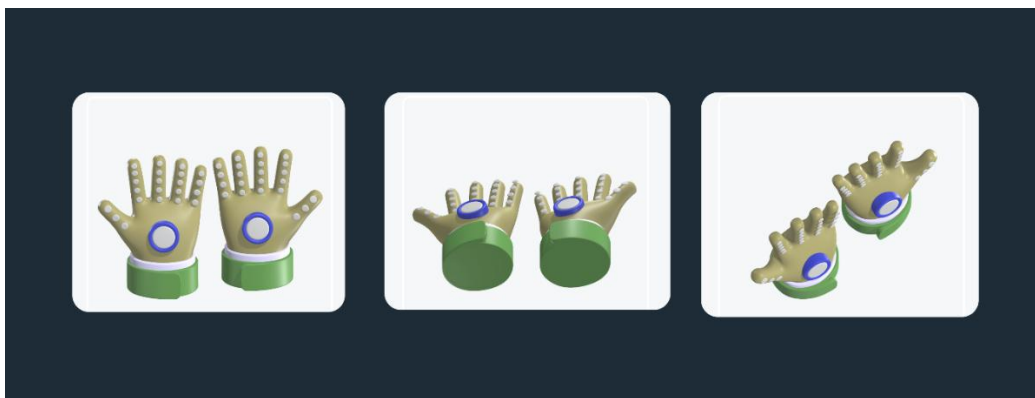


Figure 34 – Original Plans



Figure 35 – Example of flat UI icons

#### 4.1.4 Moodboards

I made three different moodboards in Figma to inspire my designs, one for the logo, one for the VR environment, and one for the Website UI.

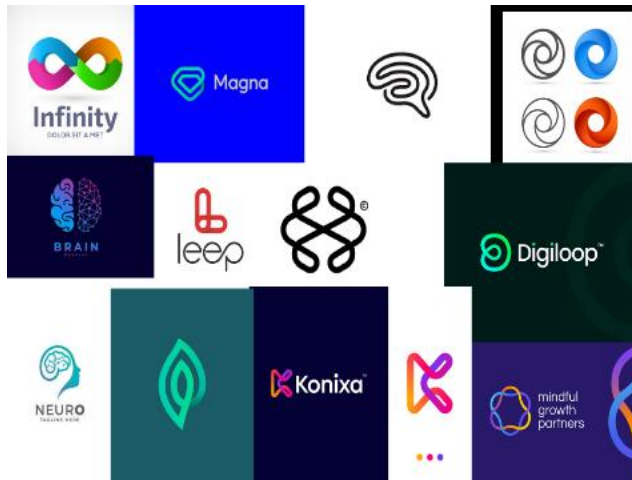


Figure 36 – Logo Moodboard



Figure 37 – VR Moodboard

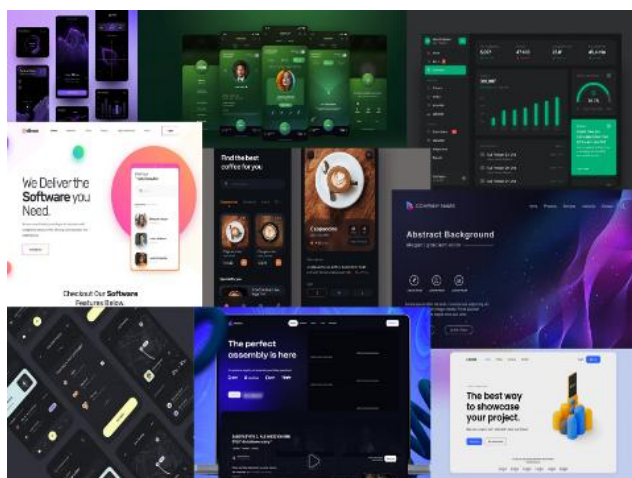


Figure 38 – Web App Moodboard

#### 4.1.5 Logo

I designed multiple logo iterations in Figma, exploring different ideas and visual styles. Initially, I was aiming for a clean, finished logo – but as the design progressed, I realised I wanted a stronger connection to VR. That led me to pivot and create a more fitting symbol, better aligned with the VR theme. After several rounds of refinement, I finalised a logo that captures both the essence of VR and the feel I was aiming for. Check out the evolution below.



Figure 39 – Early Logo Iterations

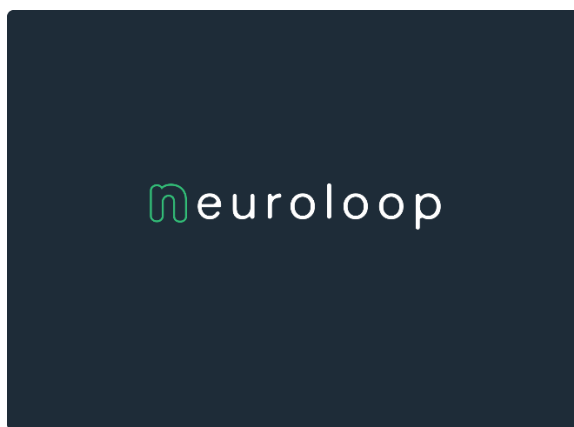


Figure 40 – Chosen Logo



Figure 41 – Final Logo

#### 4.1.6 Sound Design / Audio Feedback

Sound design and audio feedback play a vital role in enhancing user interaction within the VR application, contributing to both usability and immersion. I carefully selected and tested audio cues, and I implemented them to provide intuitive feedback for user actions, guide attention, and reinforce engagement.

For example, subtle sound effects are triggered on actions such as button clicks, successful task completions, or navigation transitions. These cues help users understand where they are and what they are doing next. Although it is just a proof-of-concept project, positive reinforcement sounds contribute to a sense of achievement and motivation to do better.

I also considered volume levels, tone of voice for the character and narrator, and noises from objects were tweaked and iterated upon as I play assessed the scenes throughout development. The spatial audio was placed so that users would not be overburdened or find the noise intrusive.

I used [Pixabay.com](https://pixabay.com) and a leading text-to-speech AI voice software by called [Eleven Labs](https://elevenlabs.com) to source and create royalty free assets and maintain quality and consistency across the platform.

Scene 1 = Log in | Scene 2 = Main menu | Scene 3 = Learn about the brain | Scene 4 = Test your knowledge (Quiz).z

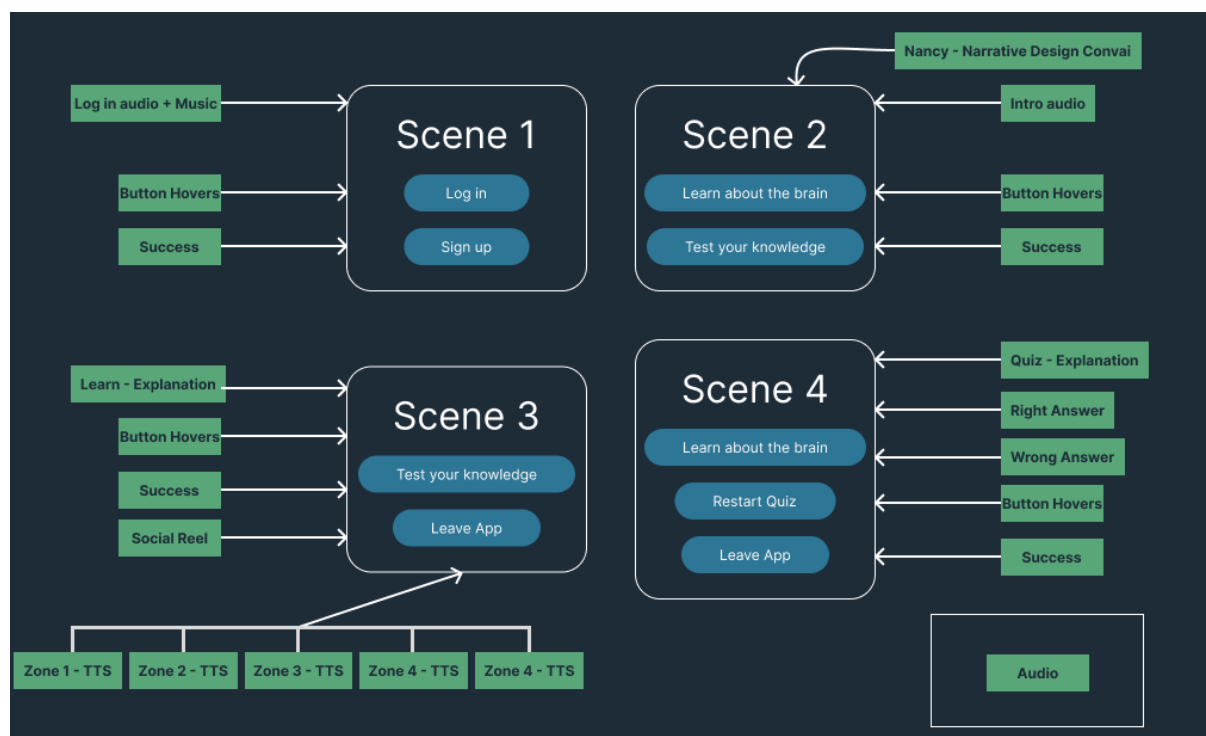


Figure 42 – Sound Design / Audio Feedback

## 4.2 Application Structure & Wireframes

### 4.2.1 Web App Architecture

This is the updated version of the website architecture. The only real way you can go wrong (that I can see) is if you do not tick 'terms & conditions,' you cannot create a profile. Otherwise, you can continue as a guest and check any of the features on the main pages.

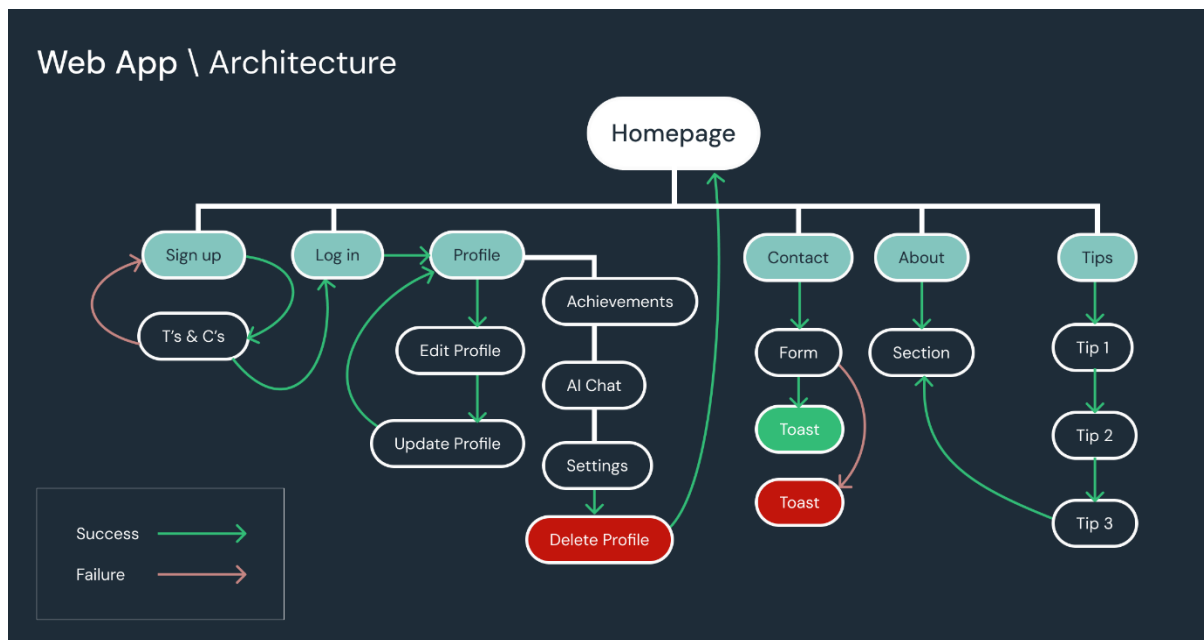


Figure 43 – Web App Architecture

## 4.2.2 Web App Wireframes

My designs outlined a fully responsive web application. I adopted a mobile-first approach, starting with layouts optimised for smaller screens to ensure a smooth and intuitive user experience. Once the mobile designs were completed, I expanded and adapted the designs for larger screen sizes, maintaining consistency and usability across devices. Below are the original Figma wireframes showing the evolution from mobile to desktop.

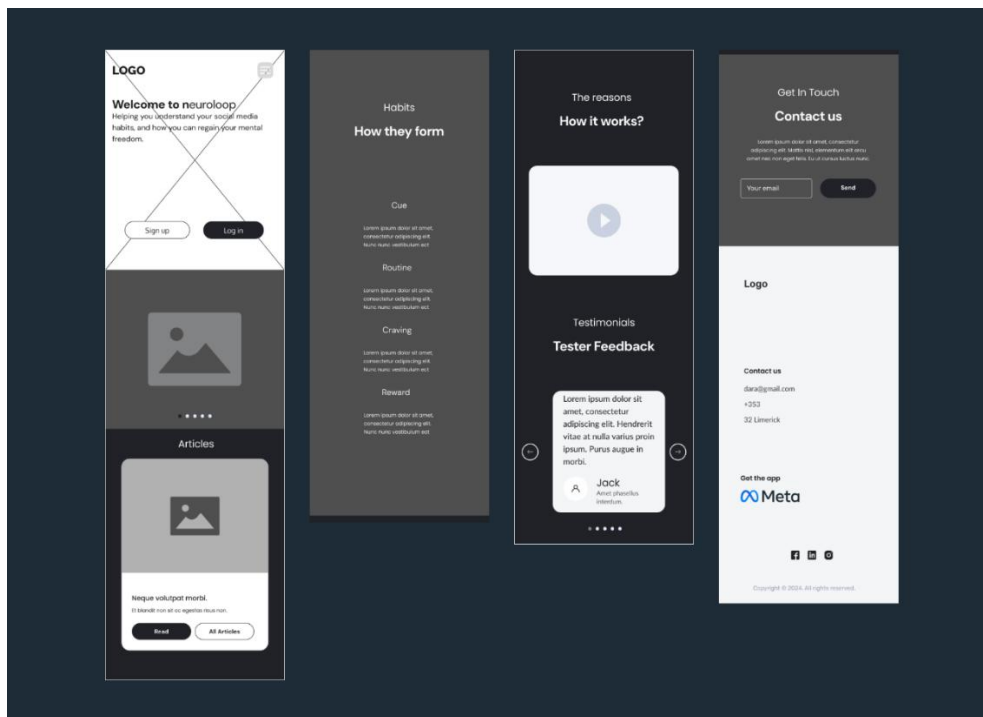


Figure 44 - Mobile homepage; Above the fold, Articles, Habits, How it works?, Test Feedback, Contact us, Footer – wireframes

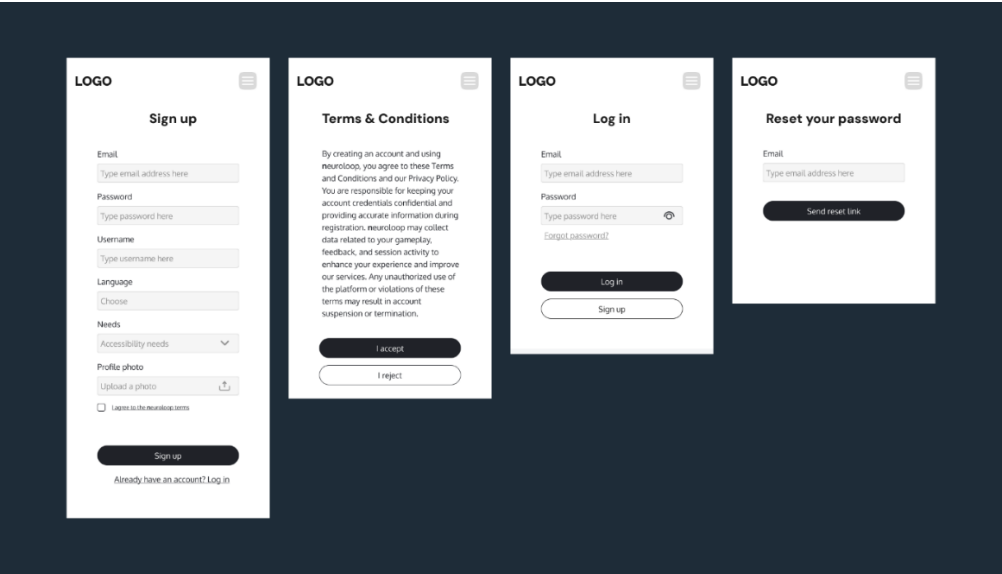


Figure 45 Sign up, T&C's, Login, Forgot Password – wireframes

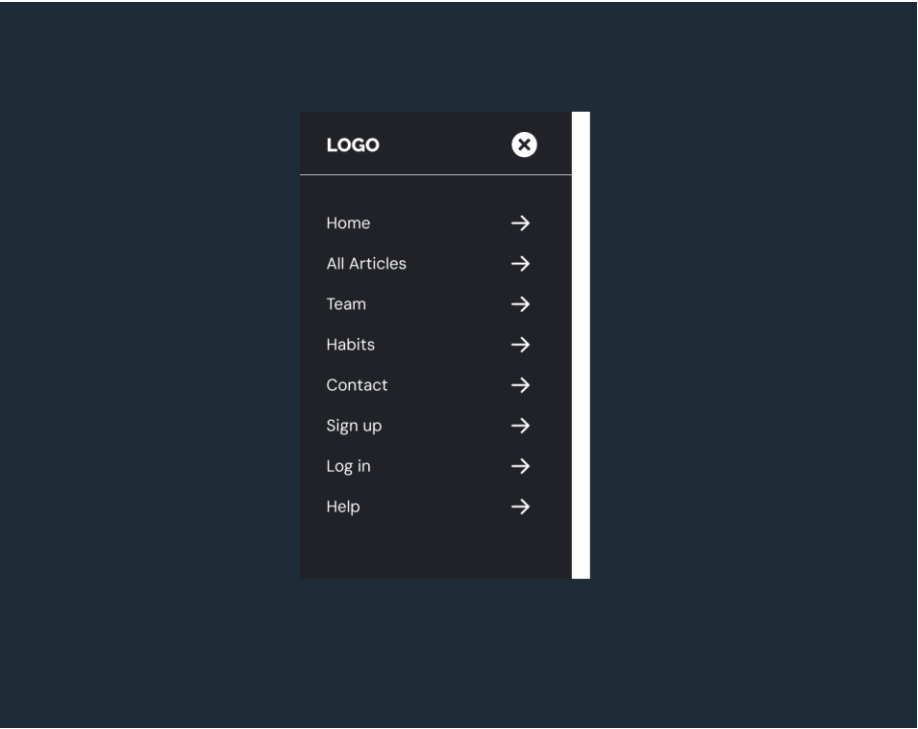


Figure 46 Hamburger menu – wireframe

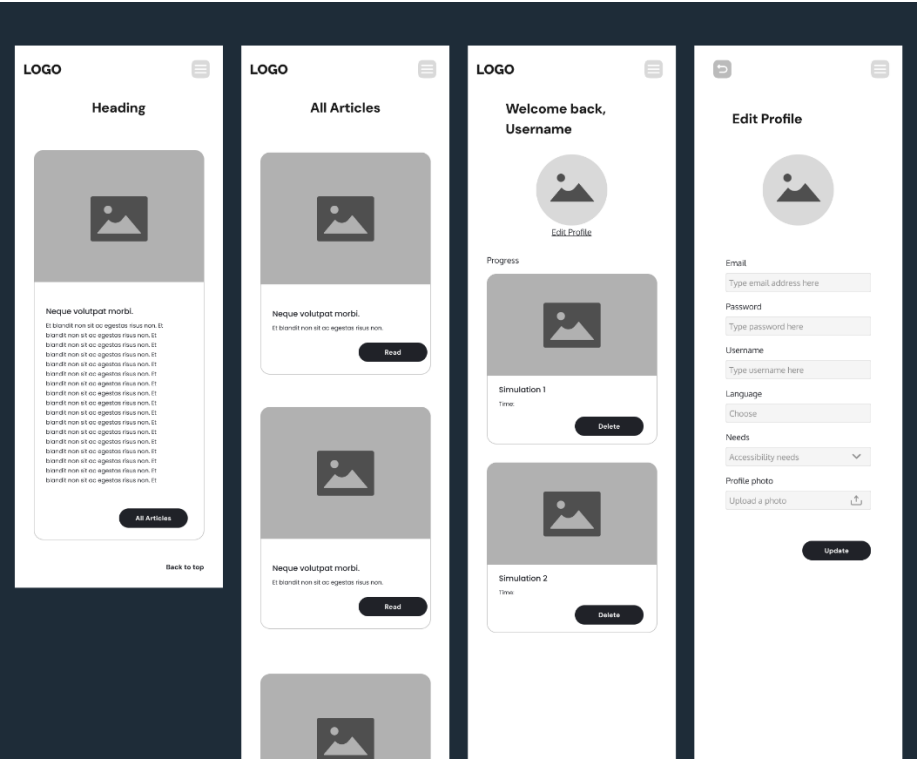


Figure 47 Read Article, All Articles, Profile, Edit Profile - wireframes

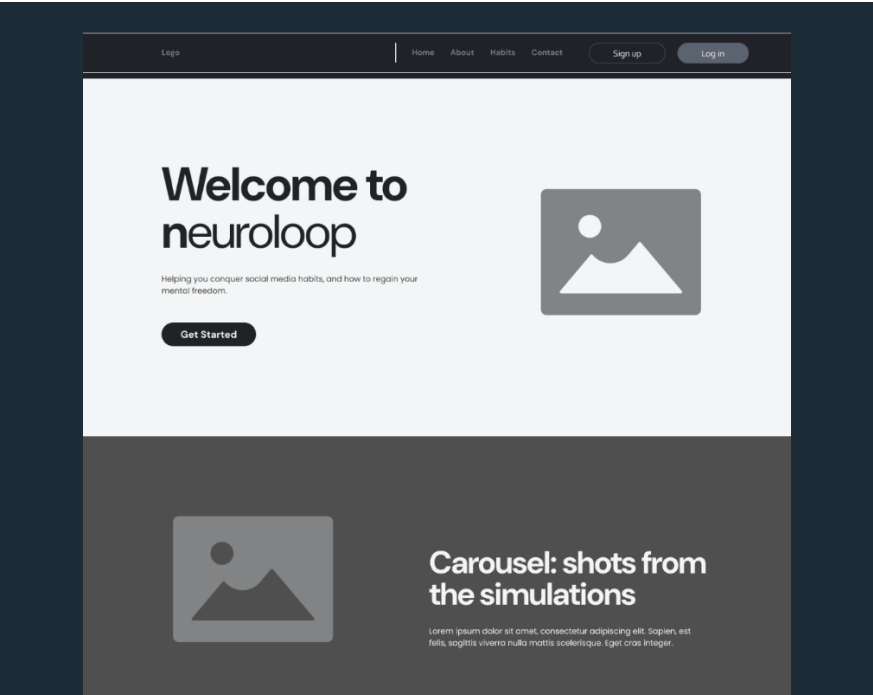


Figure 48 Desktop homepage – wireframes

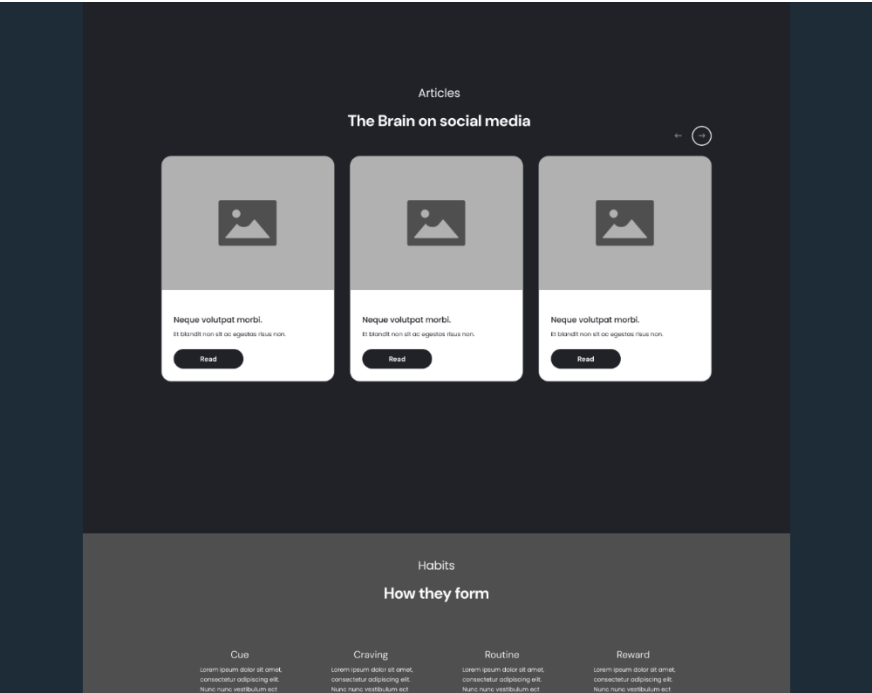


Figure 49 Desktop homepage – wireframes

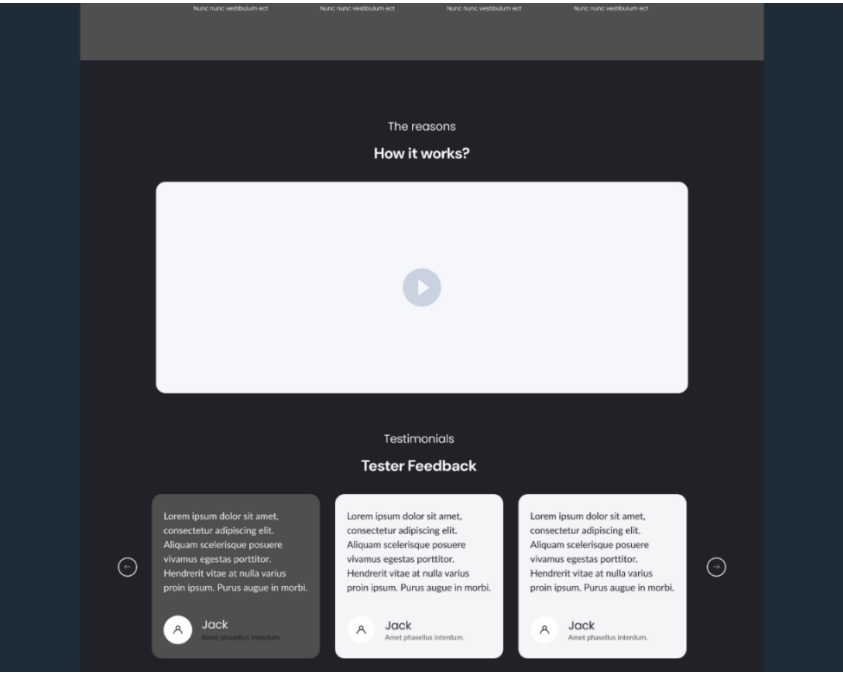


Figure 50 Desktop homepage – wireframes

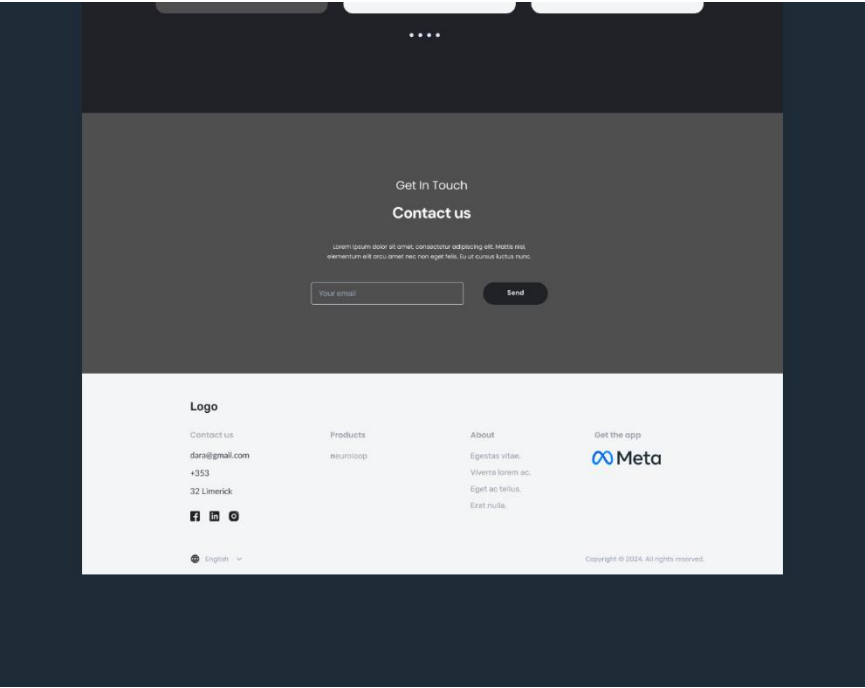


Figure 51 – User/Guest Feedback

### 4.2.3 Web App – Mid Fidelity Prototypes

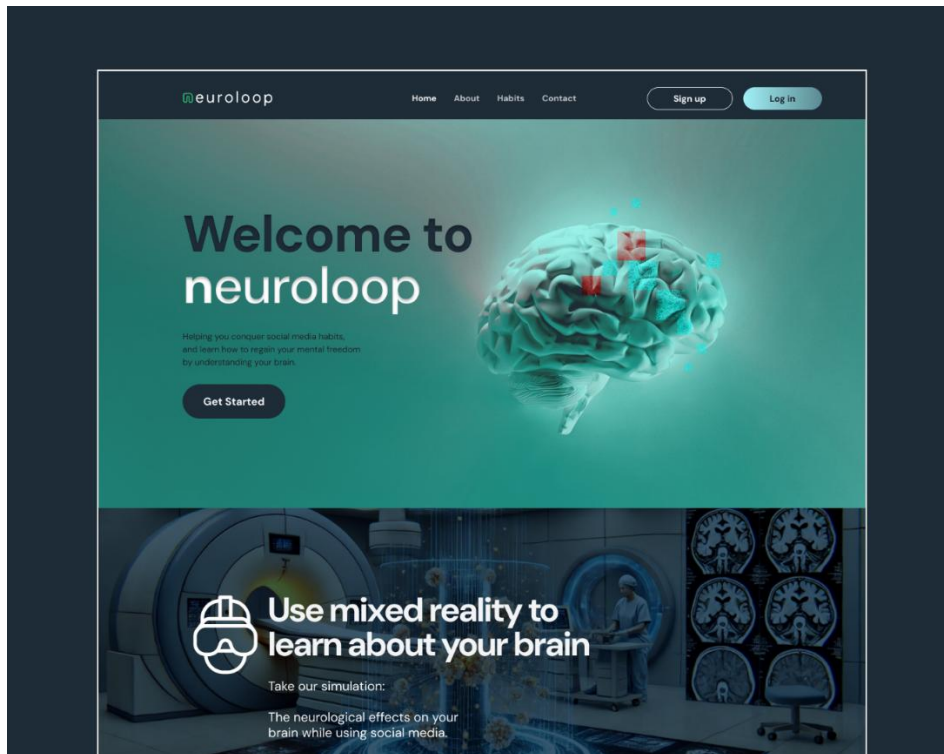


Figure 52 - Desktop Homepage - wireframe 1

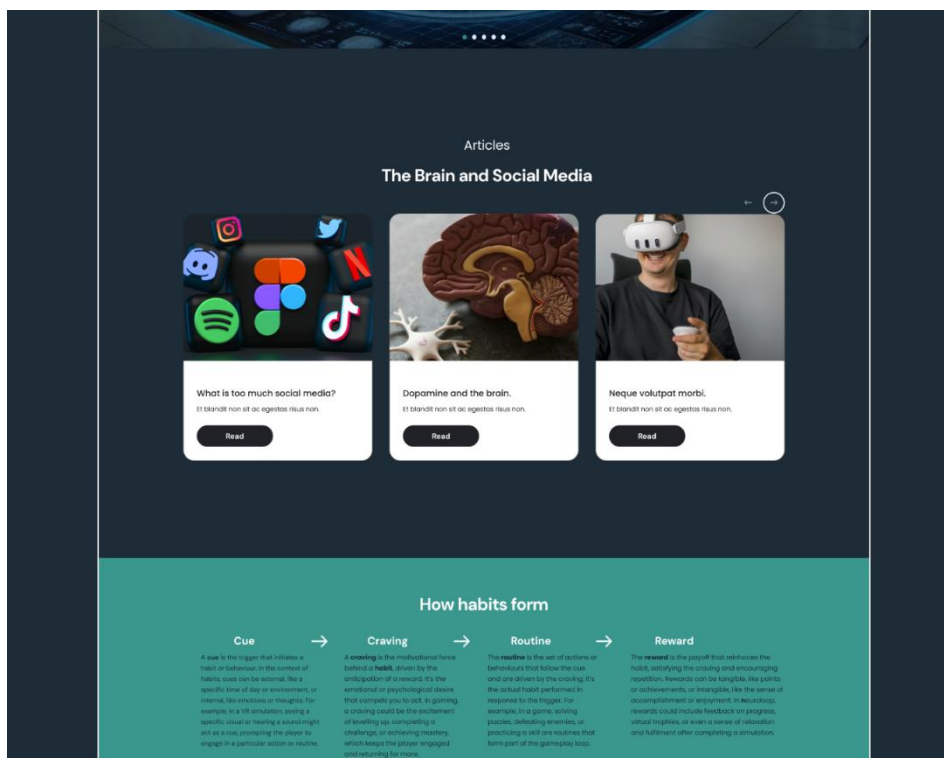


Figure 53 Desktop Homepage - wireframe 2



Figure 54 Desktop Homepage - wireframe 3

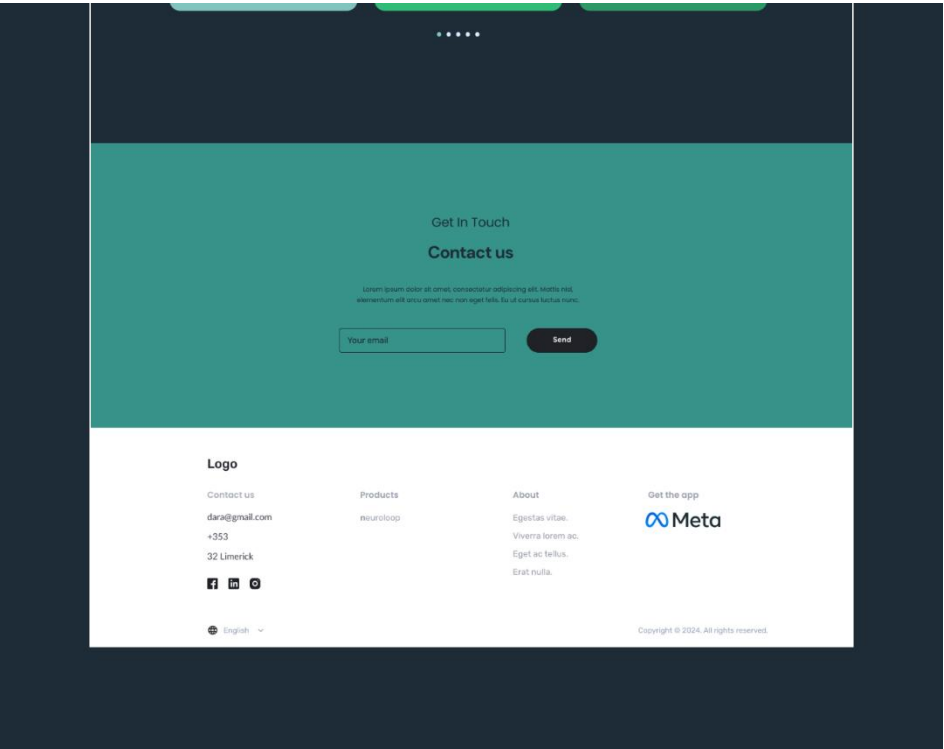


Figure 55 Desktop Homepage - wireframe 4

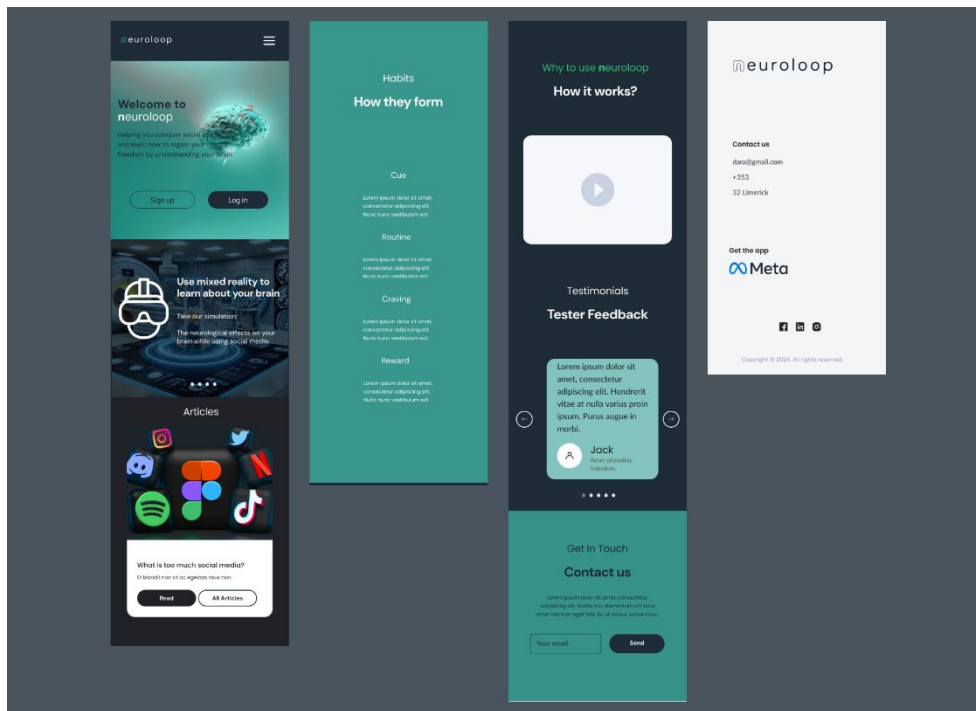


Figure 56 Mobile Homepage - Mid-fi wireframes

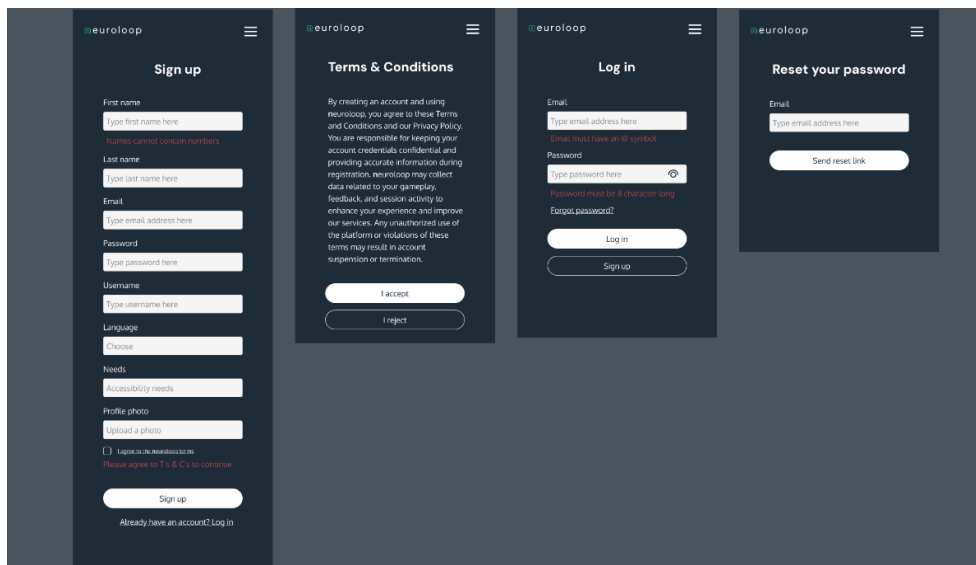


Figure 57 Mobile Signup/login with feedback messages - Mid-fi wireframes

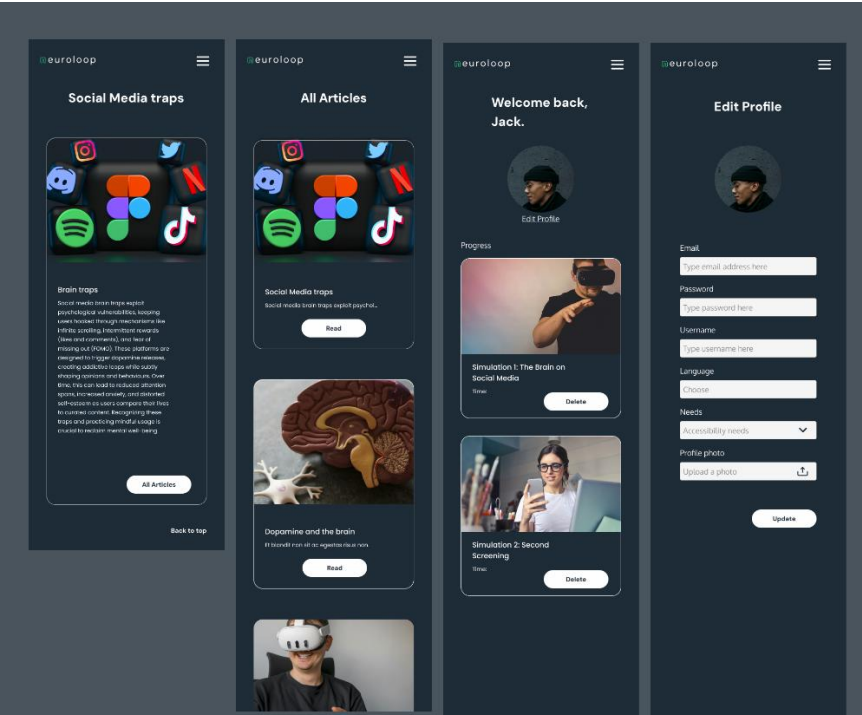


Figure 58 Mobile Articles, Profile - Mid-fi wireframes

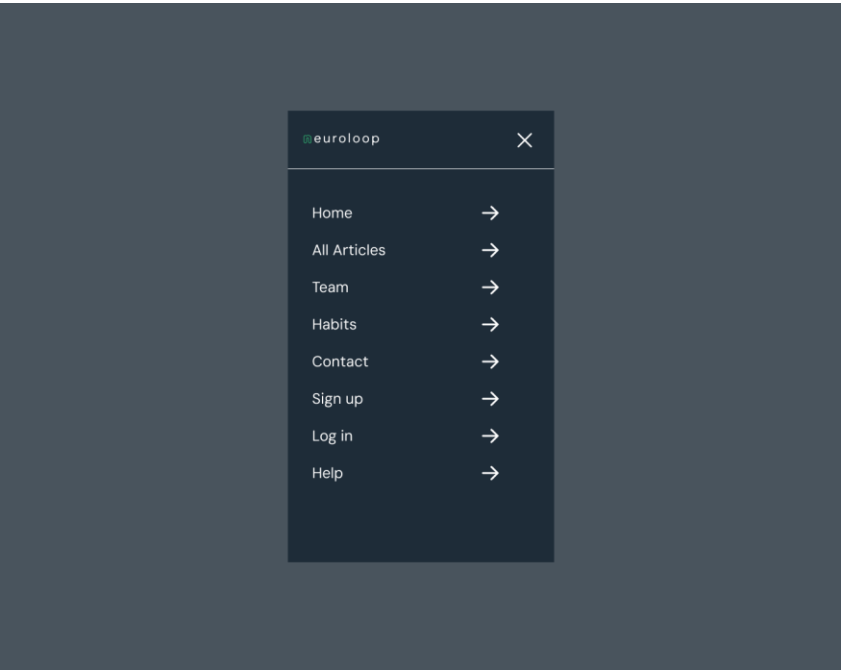


Figure 59 Mobile hamburger menu - Mid-fi wireframes

## 4.2.4 VR App Architecture

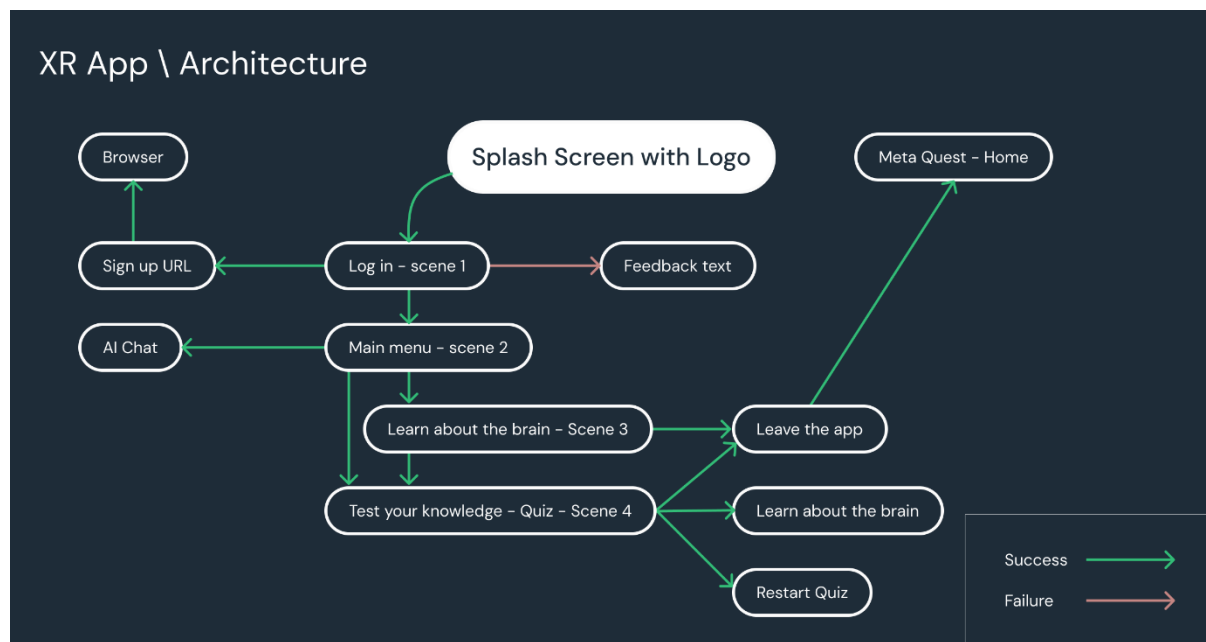


Figure 60 – VR App \ Architecture

## 4.2.5 VR Wireframes

I am not going to lie, it was extremely difficult for me to plan out how I was going to organise the scenes. The more I researched, the more possibilities I found, and it was driving me crazy to be frank. I delved into the world of using prototype apps on the Meta Quest, but it seemed quite counter-productive. I realised the most success when I put pen to paper and began to build out simple scenes in Unity. Here is a rough draft of the 'Learn about your brain' scene.

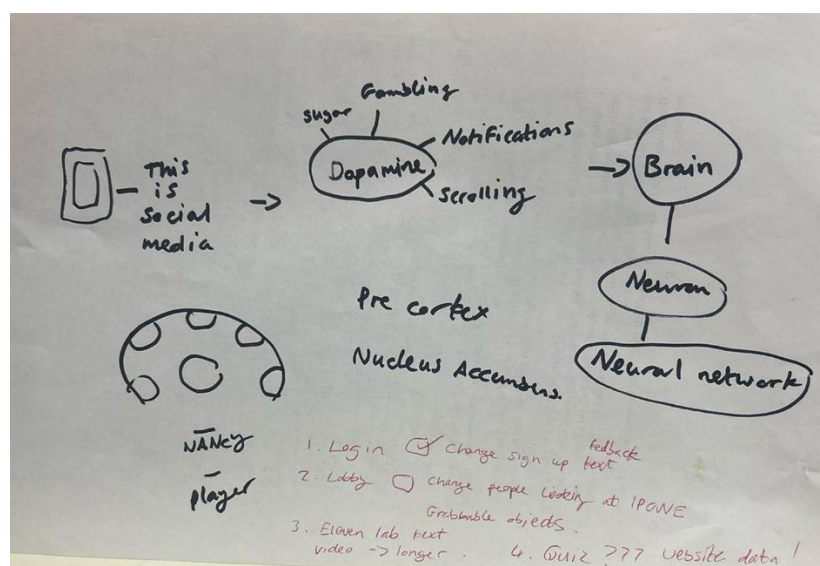


Figure 61 – VR 'Learn about your brain' scene wireframe

## 4.3 Technical Design

### 4.3.1 Project Architecture

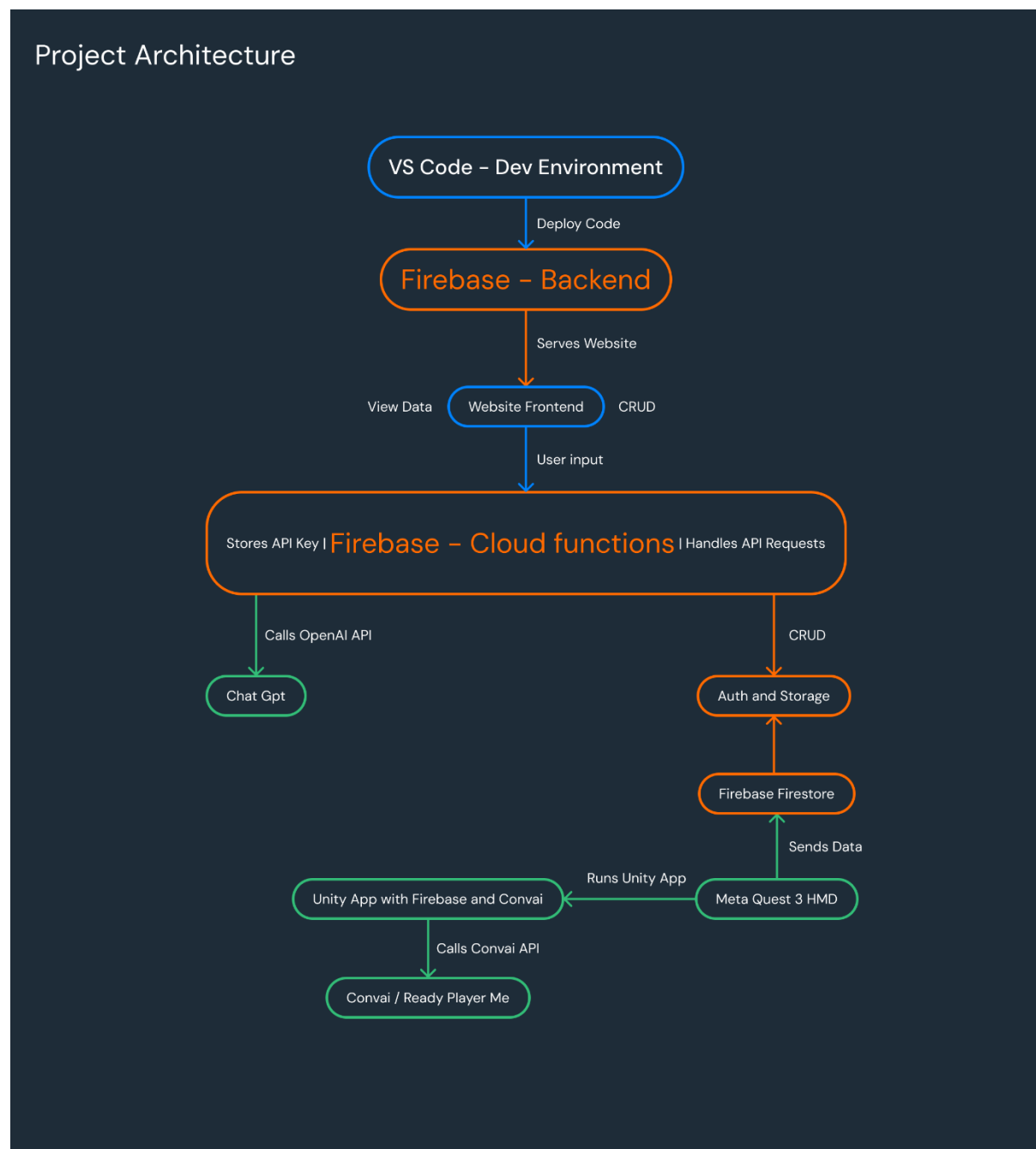


Figure 62 – Project Architecture

### 4.3.2 Third-Party Integrations

#### 4.3.2.1 ChatGPT-3.5 Turbo

Because my application as a whole is geared towards education, I wanted to provide users with access to dynamic, AI-generated content in addition to the static resources on the website. To achieve this, I integrated the OpenAI's LLM API – ChatGPT 3.5 Turbo (The most affordable model). This allowed users to ask questions directly within the platform, offering real-time support for enhancing the learning experience.

I started by reading the documentation on the OpenAI website, where I learned the necessary steps to obtain the API key and securely connect the frontend to the model. Further technical details, including implementation steps and UI integration, are covered in the Implementation section.

#### 4.3.2.2 Convai / ReadyPlayerMe

To create a more immersive and interactive VR experience, I integrated Convai for conversational AI and used ReadyPlayerMe to make the character, with whom the users could talk to if they wished. Together, these tools provided a framework for developing a responsive, life-like NPC (Non player character) that users could interact with naturally. This combination not only added personality to the environment but also aligned with the educational goals of the application by making digital learning feel more human and relatable. To make this possible, I visited the Convai website and read the documentation to learn how to get access to the API key and integrate it into my Unity project. Further technical details, including implementation steps and UI integration, are covered in the implementation section.

### 4.3.3 Backend and Database Design

For the backend infrastructure of the application, I used Firebase due to its scalability, ease of integration with frontend frameworks, and real-time capabilities. Several Firebase services were used to manage different backend functionalities:

- Firebase Authentication was implemented to handle secure user sign up/log in and session management.
- Cloud Firestore served as the primary NoSQL database, storing user data, and any dynamic content needed by the application.

- Firebase Storage – I upgraded to a paid plan to manage and media files uploaded to the website (Profile photo).
- Cloud Functions were used to securely handle the API calls to OpenAI.

This setup allowed for a robust, scalable backend without the overhead of managing a traditional server environment. It also enabled quick iteration when deploying the website.

#### 4.3.4 System Workflow

The system workflow outlines how users interact with the application and how data flows between the frontend, backend, and third-party services. When accessing the platform, users are asked to sign up and log in which is authenticated via Firebase Authentication, which manages sessions securely. Once authenticated, the application fetches the user's profile data from the sign up process via Firestore, and displays it dynamically using JavaScript.

When users update their profile information or upload a profile photo, the data is validated on the client side, then sent to Firestore and Firebase Storage. Images that have been uploaded are handled asynchronously, and download URLs are retrieved to update the user's document in the database.

For interactive support, users can access the ChatGPT-3.5 Turbo UI on their profile. Their chat inputs are sent to Firebase Cloud Functions, which securely handle the request and fetch a response from the OpenAI API, ensuring that the API key is not exposed on the client side.

In the VR application, Convai powers a conversational NPC – Nancy (Created using RPM). These tools are integrated into the Unity environment, creating a seamless blend between traditional web content and immersive 3D interactions.

This workflow provides a modular, secure, and user-friendly experience, combining real-time database operations, AI-enhanced interactivity, and immersive VR elements into a cohesive educational platform.

## 5 Implementation

### 5.1 Environment and Scene Design

The environment and scene design were crucial in creating an immersive and engaging user experience in the VR application. My primary objective was to make a virtual space that felt inviting and aligned with the educational goals of the platform.

I began by organising my setup in Unity. I created a scene and started structuring the scene component category headings to keep it organised; XR setup, Audio, Scripts, Static, Dynamic, etc. I then copied the scene and duplicated it out, and labelled the scenes accordingly, scene 1 – Log in, scene 2 – Main Menu, etc.

During the research stage of the project, I had been looking into diverse ways to speed up my workflow. By scouring the Unity Asset Store and SketchFab, I found assets that matched what I wanted to portray in my scene designs for the zones. I used blender to make assets that I couldn't find. By following tutorials, I developed the UI myself and made use of the c sharp scripts that were already present in the VR template that that I was using. For instance, to make the Log in menu follow the player and other such scripts. Lighting was configured using baked and real-time techniques to ensure realistic shadows while optimizing performance for the Meta Quest HMD. Spatial sound cues were also implemented to enhance user orientation and engagement within the environment. Below you will find screenshots of gameplay and my Unity IDE.

Scene 1 – Log in UI - I tried to keep it uniform with my website (icons, colours, font). There's a link to sign up which opens a browser window in this scene and a script to hide and show your password using the eye open/closed icon.

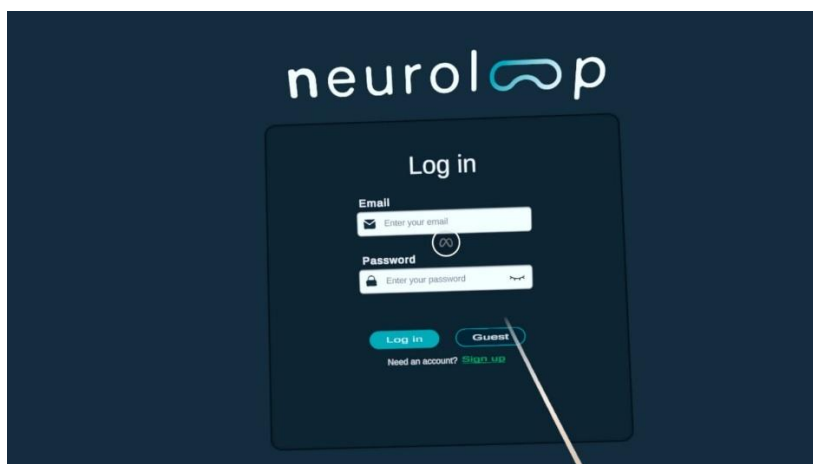


Figure 63 – Scene 1

Scene 2 – Logged in - A tutorial for the controllers, and UI provided by the template to allow more than one user to be in the same scene from different headsets.



Figure 64 Scene 2 – Instructions

Scene 2 – Logged in - Corridor leading up to the main menu and Convai character – Nancy.

I wanted this to provide anticipation and give the user a chance to learn how to use the different forms of locomotion (movement) in the app.

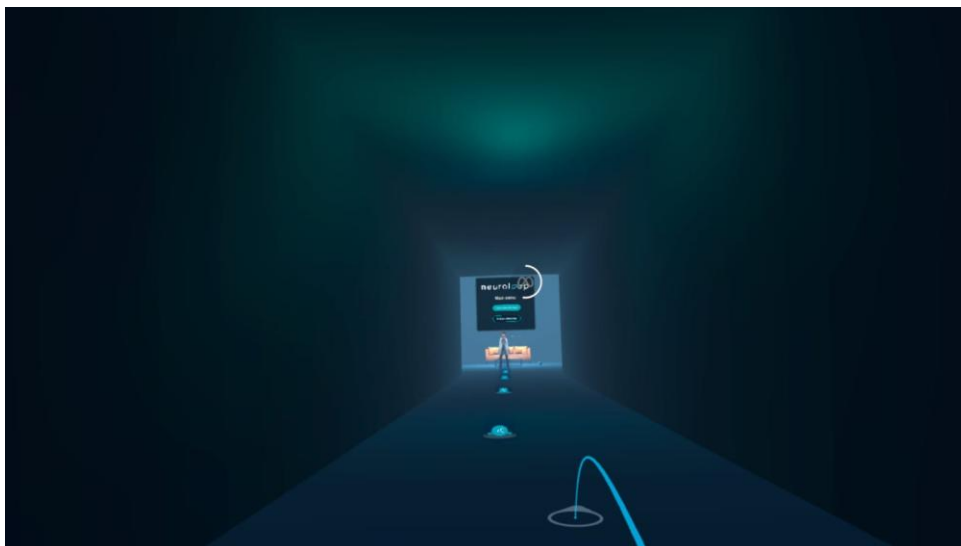


Figure 65 – Corridor for anticipation

Scene 2 – Logged in - Nancy with the UI panel showing what the user says and what her response is. The main menu is behind her along with a few interactable items.



Figure 66 – Convai UI and Main Menu UI

Scene 3 – Learn about the brain – Zone 3, a labelled neuron interactable and synapse animation.

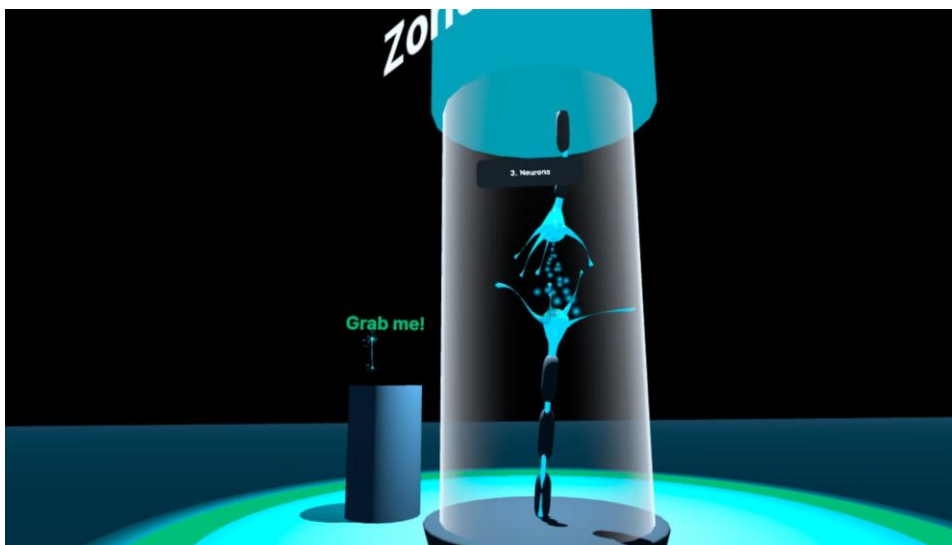


Figure 67 – Learn about the brain - Zone 3

Scene 3 – Learn about the brain – Zone 1, a Large mobile phone with an animation, and a small interactable phone displaying an instagram reel with audio.

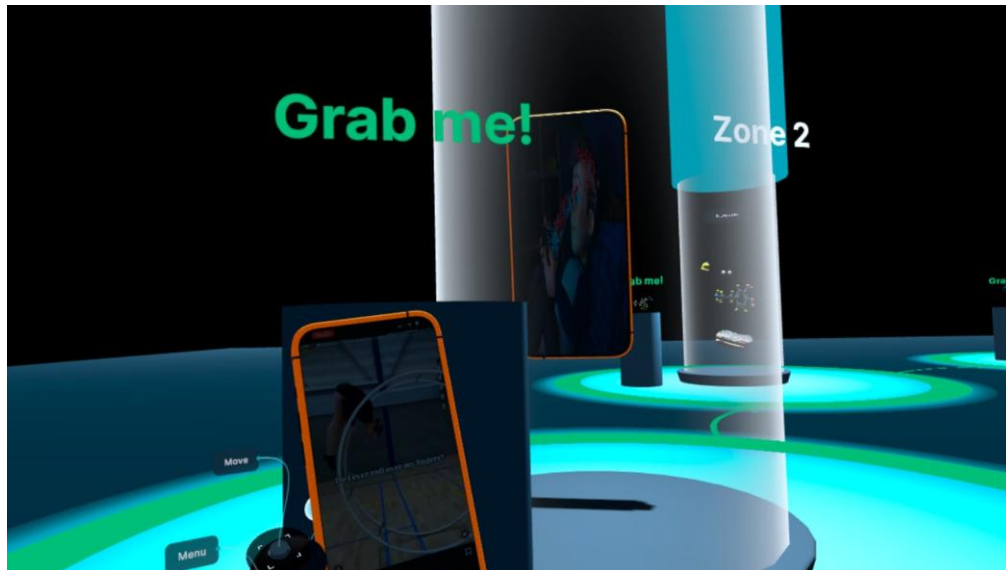


Figure 68 – Zone 1

Scene 4 – Test your knowledge – Quiz UI that runs through ten questions about the brain, dopamine, and other relatable content.

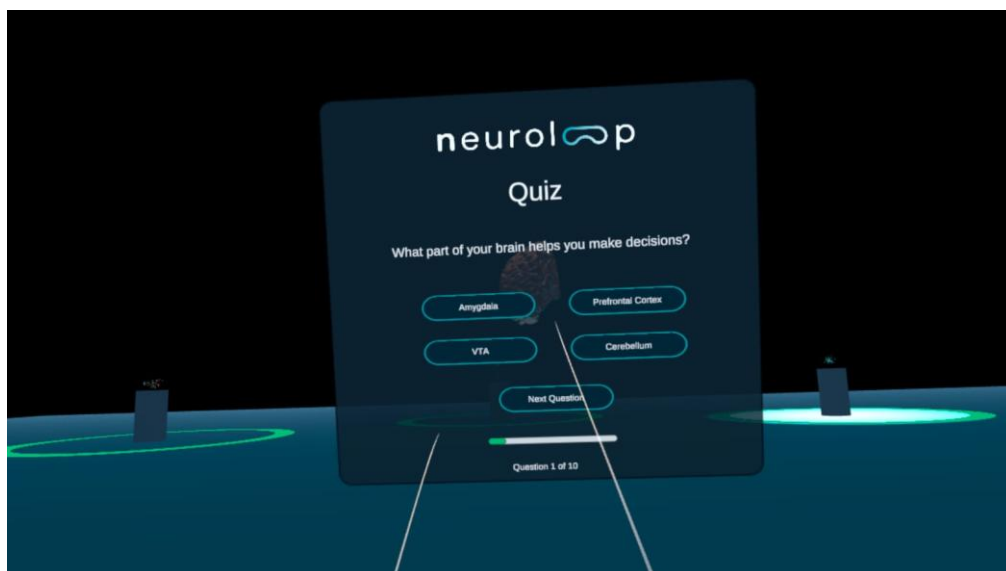


Figure 69 – Quiz UI

This is the Unity IDE. It's a little difficult to see but the Unity icons are displaying my scenes that are stored in my myScenes folder. The hierarchy to the left shows the structure of my scene elements.

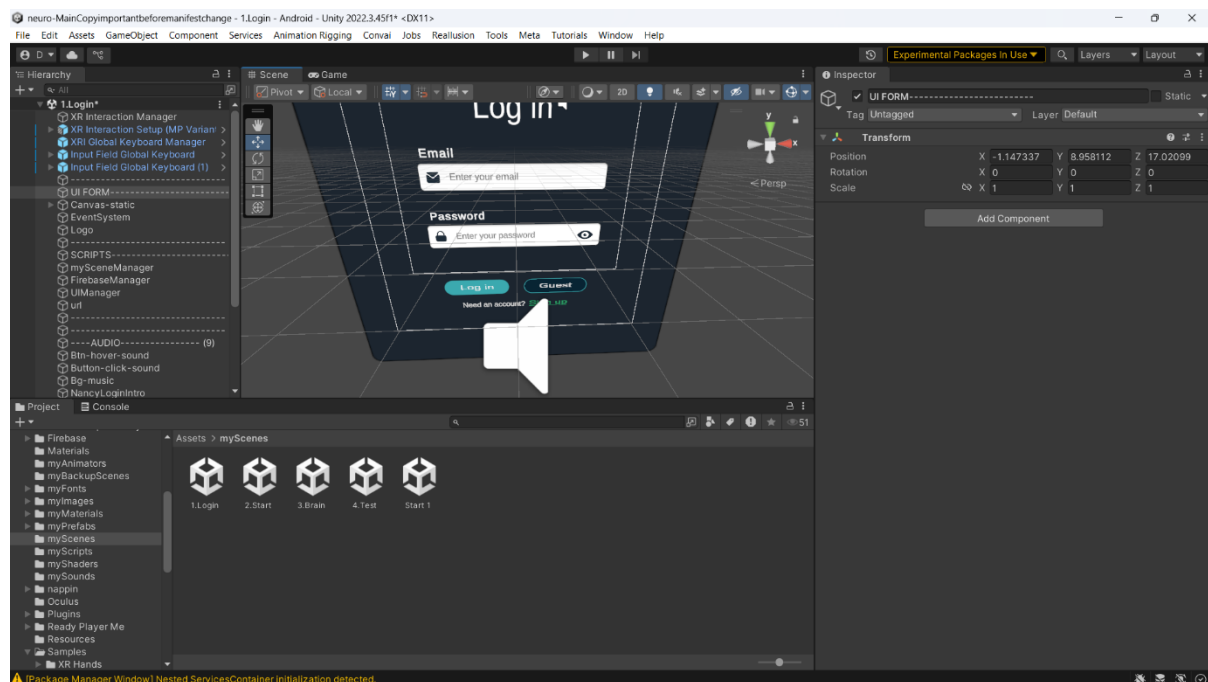


Figure 70 – Unity IDE – Scene Design

This is side view of scene 2 – I created the dome asset in blender and imported it into my project to act as the room.

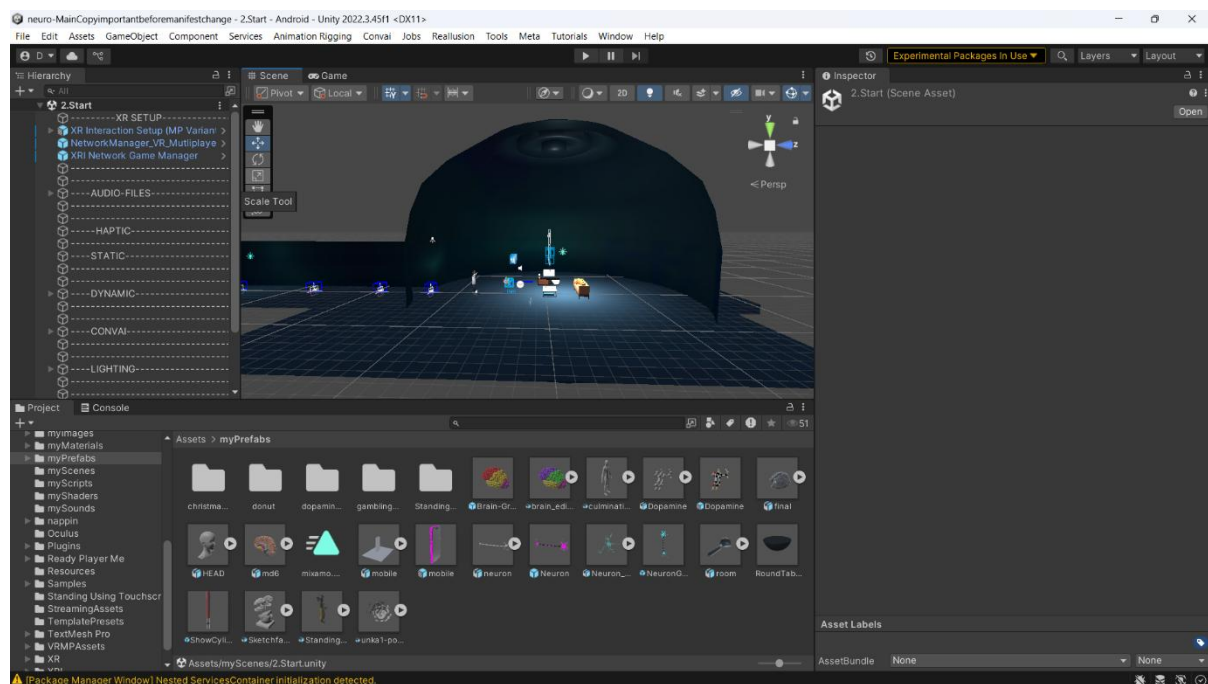


Figure 71 – Scene 2 – Blender Asset Example



## 5.2 Interactivity and User Flow

Interactivity and user flow were central to the design in the VR experience. It was important to ensure users could intuitively navigate the application, engage with content, and achieve their goals with unnecessary friction.

The user journey as shown above, is quite clearly marked out. They have the option to roam freely, but I tried my best to make sure that they always knew what they were doing, and what they needed to do next. Because it is a VR application, there is a level of learning and time spent in the application to figure out movement, or even get your head around being in a virtual space. There is a player menu that allows you to control your movement settings and make yourself as comfortable as possible.

## 5.3 Voiceover and Audio Integration

Audio was a key component in enhancing immersion and engagement. The integration of both ambient sound and voiceovers contributed to a more realistic and emotionally engaging experience. The music for the Log in was carefully chosen to be uplifting and give a sense of adventure.

For voice interactions, Convai was used to power real-time conversational AI for the NPC. The character responds to users with synthesized speech, creating the illusion of live dialogue. On the Convai website, when developing 'Nancy,' I inputted everything that she knew, but I also chose her tone, and created a narrative design that allowed me to trigger a speech when the user first approaches her using an empty game object and a script.

I used voiceovers to explain what to do next and to relay the educational information. Volume levels, timing, and sound priority were carefully balanced to avoid overwhelming the user, ensuring that voiceovers and ambient sounds worked together to support the educational goals of the application. The spherical lines in the figure below depict the spatial reach of the audio objects. When a user steps into a zone (green circle) the voiceover is triggered to play.

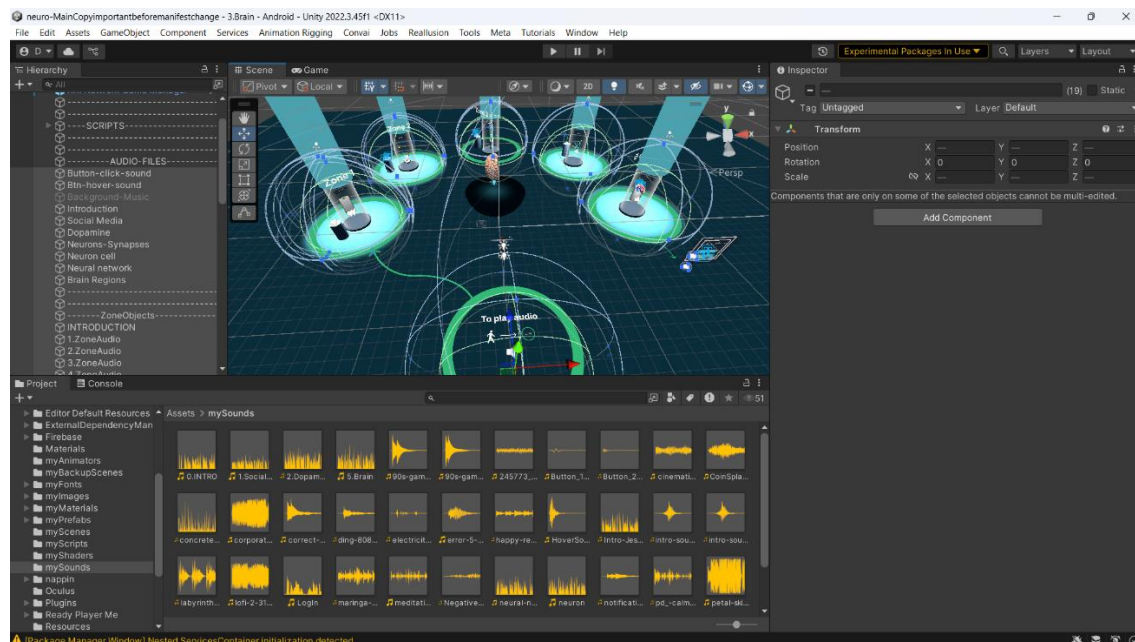


Figure 74 – Audio Integration

## 5.4 Technical Implementation

### 5.4.1 Overview

- Firebase (Auth, Storage, Firestore, Functions)
- Unity (VR environment, Scripting)
- VS Code (JavaScript frontend logic)
- Convai (NPC voice interaction)
- OpenAI API (ChatGPT)

See [Figure 62 – Project Architecture](#)

## 5.4.2 Code Integration

### 5.4.2.1 Web Application

This section shows how I integrated Firebase into my website using - VS Code.

#### 5.4.2.1.1 Firebase

Firebase was instrumental in sewing my application together as whole. I read the documentation and followed the steps for implementation. For anything that was confusing or causing issues I followed YouTube tutorials and read from online forums to distinguish versioning and solve any hiccups I encountered.

##### 5.4.2.1.1.1 Authentication

Initialising Firebase in my project.

```
public > assets > js > JS firebase.js > ...
1  import { initializeApp, getApps } from "https://www.gstatic.com/firebasejs/11.1.0/firebase-app.js";
2  import { getAuth } from "https://www.gstatic.com/firebasejs/11.1.0/firebase-auth.js";
3  import { getDatabase, ref, onValue } from "https://www.gstatic.com/firebasejs/11.1.0/firebase-database.js";
4  import { getStorage } from "https://www.gstatic.com/firebasejs/11.1.0/firebase-storage.js";
5  import { getFirestore } from "https://www.gstatic.com/firebasejs/11.1.0/firebase-firestore.js";
6
7  const firebaseConfig = {
8    apiKey: "AIzaSyDz3neRTIE6-0MXZEsdR1w7-jCo3-6ysiq",
9    authDomain: "neuroloop-13690.firebaseio.com",
10   databaseURL: "https://neuroloop-13690-default-rtdb.europe-west1.firebaseio.com",
11   projectId: "neuroloop-13690",
12   storageBucket: "neuroloop-13690.appspot.com",
13   messagingSenderId: "350302613612",
14   appId: "1:350302613612:web:2f01b07d3261df64ef6849",
15   measurementId: "G-66KVPQ3H27"
16 };
17
18 // Initialise Firebase
19 const app = !getApps().length ? initializeApp(firebaseConfig) : getApps()[0];
20
21
22 const auth = getAuth(app);
23 const database = getDatabase(app);
24 const storage = getStorage(app);
25 const firestore = getFirestore(app);
26
27
28 export { auth, database, storage, firestore, ref, onValue };
29
```

Figure 75 – Initialising Firebase

Firebase Auth was used to manage user sign-up, login, and persistence across the platform. This code block initialises Firebase with the project configuration from the Firebase console. Captures the login form input (email and password). Validates the inputs. Uses **firebase.auth().signInWithEmailAndPassword()** to sign in the user. If successful, the user is redirected to their profile page. If not, the error message is displayed.

```

43
44 // Firebase Authentication: Sign in user
45 firebase.auth().signInWithEmailAndPassword(email, password)
46 .then((userCredential) => {
47   const user = userCredential.user;
48   console.log('User logged in:', user);
49
50   // Redirect to profile or home page after successful login
51   window.location.href = "/profile.html"; // Or wherever you want to send the user
52 })
53 .catch((error) => {
54   const errorCode = error.code;
55   const errorMessage = error.message;
56
57   // Handle errors and show appropriate messages
58   console.error(errorCode, errorMessage);
59
60   // Display error message on the screen
61   if (errorCode === 'auth/user-not-found' || errorCode === 'auth/wrong-password') {
62     alert('Incorrect email or password. Please try again.');
```

Figure 76 - Firebase Authentication

#### 5.4.2.1.1.2 Functions

In this block of code, the Cloud Function (**getOpenAIResponse**) acts as a secure backend proxy to interact with OpenAI API (ChatGPT-3.5 Turbo). Firebase's `onRequest` trigger responds to Http requests. It loads the OpenAI API key securely using **defineSecret()** so it's not exposed on the frontend. It enables CORS (Cross-Origin-Resource-Sharing), allowing requests from the website. Validates the incoming requests to ensure it includes a valid prompt. Sends the user's prompt to the OpenAI Chat Completion API. Returns the AI response to the front end.

```

const OPENAI_API_KEY = defineSecret("OPENAI_API_KEY");

exports.getOpenAIResponse = onRequest(({
  region: "europe-west1",
  secrets: [OPENAI_API_KEY],
}), (req, res) => {
  cors(req, res, async () => {
    const openaiKey = OPENAI_API_KEY.value();

    if (!openaiKey) {
      logger.error("Missing OpenAI API key.");
      return res.status(500).send("Missing API key.");
    }

    if (!req.body.prompt || typeof req.body.prompt !== 'string') {
      return res.status(400).send("Invalid or missing prompt.");
    }

    try {
      const response = await axios.post(
        "https://api.openai.com/v1/chat/completions",
        {
          model: "gpt-3.5-turbo",
          messages: [{ role: "user", content: req.body.prompt }],
          max_tokens: 100,
        },
        {
          headers: {
            Authorization: `Bearer ${openaiKey}`,
            "Content-Type": "application/json",
          },
        }
      );
    }
  });
});
```

Figure 77 – Firebase Functions Get Open AI Response

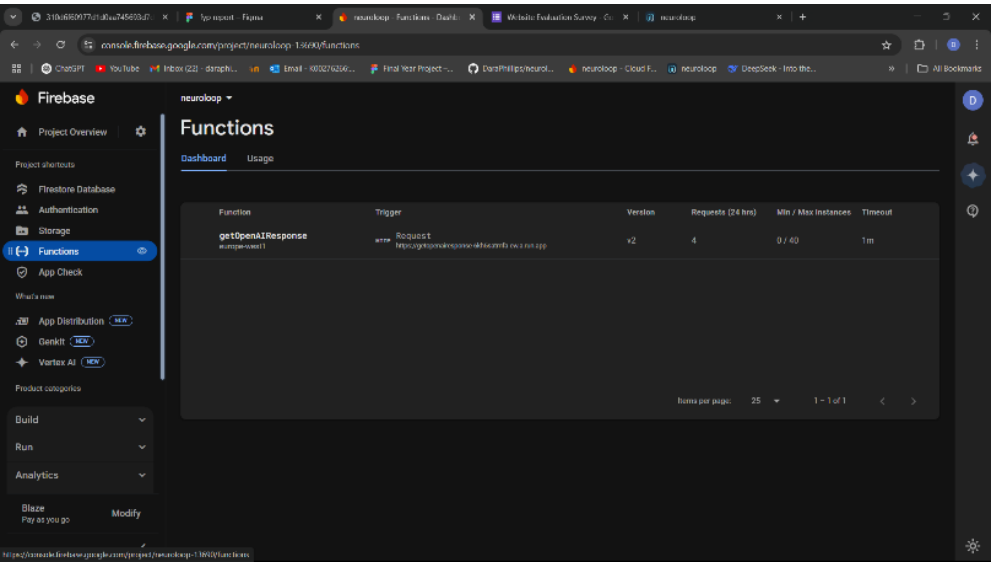


Figure 78 – Firebase Functions getOpenAIResponse

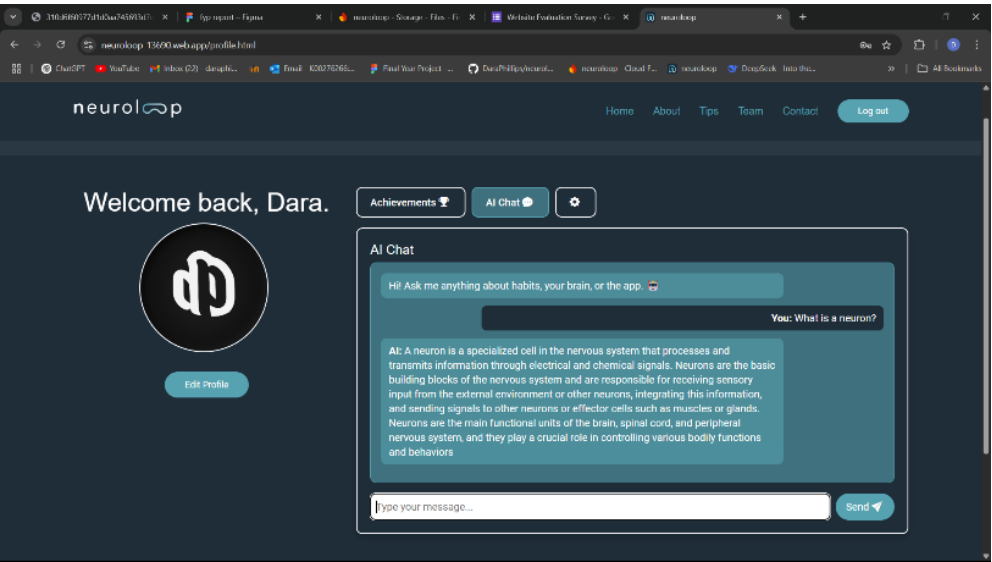


Figure 79 – ChatGPT-3.5 Turbo UI

### 5.4.2.1.1.3 Firestore

I used Firestore to manage sign-up, update, and for feedback form submissions.

When a user signs up, their credentials are authenticated using Firebase Auth. Additional user information, such as name and email, first name, last name etc are stored in Firestore using **setDoc ()** under the user's unique ID which it also automatically assigns.

```
photoURL = await getDownloadURL(storageRef);
}

await setDoc(doc(db, "users", userId), {
  firstName,
  lastName,
  email,
  username,
  termsAccepted: true,
  profilePhoto: photoURL,
  createdAt: new Date().toISOString(),
});
```

Figure 80 – Firestore Sign up Data

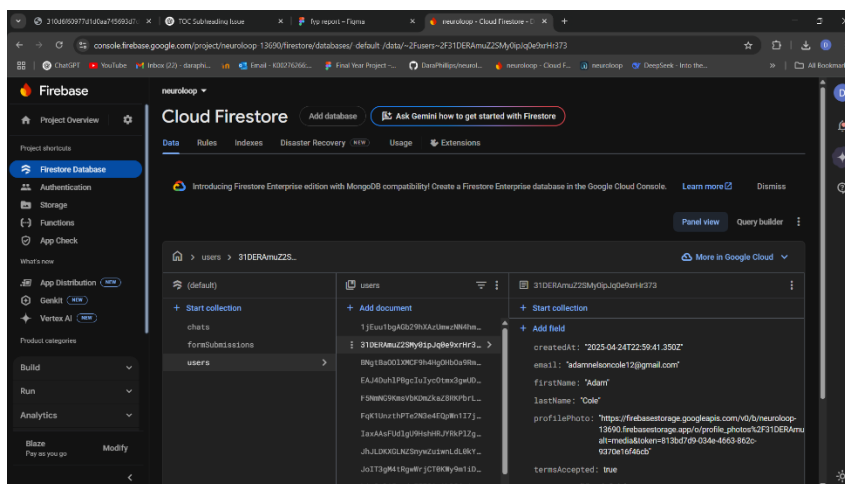


Figure 81 – Firestore Documents Console

When updating user profiles, Firestore uses **setDoc()** and the **{ merge: true }** option. This ensures that only specified fields are updated without affecting other user data already stored.

```

const userData = {
  ...(firstName && { firstName }),
  ...(lastName && { lastName }),
  ...(username && { username }),
  ...(photoURL && { profilePhoto: photoURL }),
  updatedAt: new Date().toISOString(),
};

await setDoc(doc(db, "users", user.uid), userData, { merge: true });
alert("Changes saved successfully.");
window.location.href = "/profile.html"; // <-- Redirect here
} catch (err) {
  console.error("Error saving changes:", err);
  alert("An error occurred while saving changes. Please try again.");
}
});

```

Figure 82 – Firestore Edit profile Data

To complete the form submission, Firestore uses **addDoc()** to create an entirely new document. It includes the fields set name, email, subject etc and assigns a unique ID or **uid**.

```

19
20 const name = form.name.value.trim();
21 const email = form.email.value.trim();
22 const subject = form.subject.value.trim();
23 const message = form.message.value.trim();
24
25 try {
26   // Submitting form data to Firestore
27   await addDoc(collection(firestore, 'formSubmissions'), {
28     name,
29     email,
30     subject,
31     message,
32     timestamp: new Date()
33   });
34
35   // Reset the form and show success toast
36   form.reset();
37   form.classList.remove('was-validated');
38   toast.show();
39 } catch (error) {
40   console.error('Error submitting to Firestore:', error);
41   alert('Failed to send message. Please try again.');
```

Figure 83 – Firestore Form Submission

#### 5.4.2.1.1.4 Storage

Firebase Storage is used to upload and manage the users' profile photos. Files are stored securely in Google Cloud Storage, and a unique download URL is retrieved using **getDownloadURL ()** to display in the app.

```

65
66   uploadTask.then((snapshot) => {
67     console.log('File uploaded successfully!', snapshot);
68
69     // Get the download URL
70     getDownloadURL(snapshot.ref).then((downloadURL) => {
71       console.log('File available at:', downloadURL);
72       // Store the download URL to the database after successful upload
73       saveUserData(downloadURL);
74     });
75   }).catch((error) => {
76     console.error('Error uploading file:', error);
77   });
78

```

Figure 84 – Firestore Storage Code

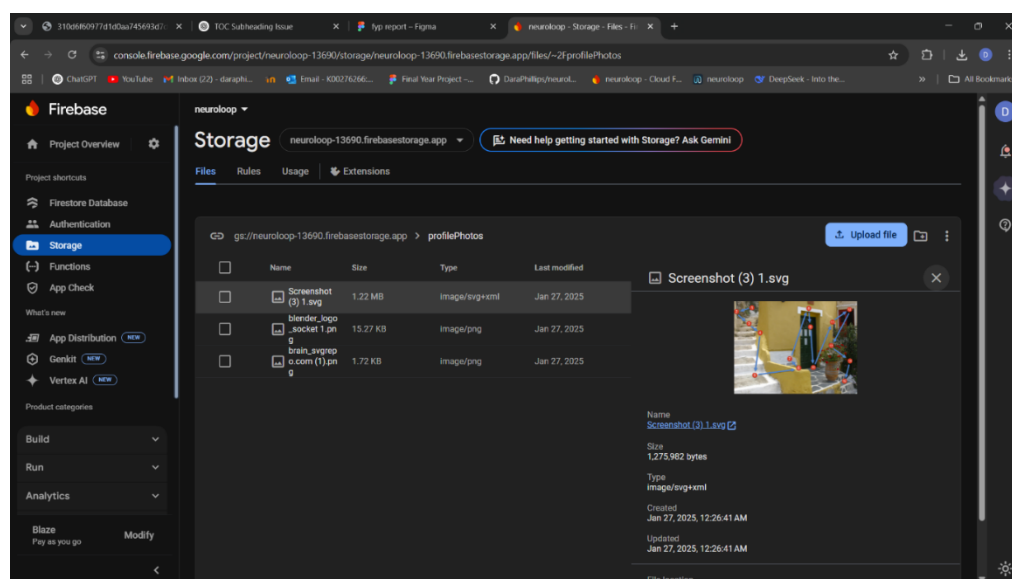


Figure 85 – Firebase Storage Console

#### 5.4.2.1.2 JavaScript Logic Examples

JavaScript was also essential to providing a dynamic aspect to my Web app. Here are some examples of the functions used to check if the user is logged in, populate the data for the profile, delete data, logout, display the levels they completed.

**OnAuthStateChanged()** Listens for changes in the users' authentication status. If a user is logged in, it fetches their data from Firestore and populates the profile UI, otherwise it redirects them to the login page. As you can see, it's also calculating the user's score from the quiz and calculating the percentage to be displayed along with the other information.

```
// Handle auth state change and get user data
onAuthStateChanged(auth, async (user) => {
  if (user) {
    try {
      const userDocRef = doc(db, "users", user.uid);
      const userDoc = await getDoc(userDocRef);

      if (userDoc.exists()) {
        const userData = userDoc.data();
        const firstName = userData.firstName || "User";
        const profilePhotoUrl = userData.profilePhoto || "/assets/img/avatar2.svg";
        const score = userData.score || 0; // Get the score from Firestore

        // Calculate the percentage out of 10
        const maxScore = 10;
        const percentage = (score / maxScore) * 100;

        // Display user information
        document.getElementById("profileName").textContent = firstName;
        document.getElementById("profilePhoto").src = profilePhotoUrl;

        // Display the score and the percentage in the profile
        document.getElementById("userScore").textContent = `Score: ${score} / ${maxScore}`;

        // Display retry message if score is below 40%
        const retryMessage = document.getElementById("retryMessage");
        if (percentage < 40) {
          retryMessage.textContent = "Your score is below 40%. We recommend retrying the quiz.";
          retryMessage.style.color = "red"; // Highlight retry message
        } else {
          retryMessage.textContent = "";
        }
      }
    } catch (error) {
      console.error("Error fetching user data: ", error);
    }
  }
});
```

Figure 86 – OnAuthStateChanged() - JS

**logoutButton.addEventListener()** This event listener logs the user out when the logout button is clicked. It calls Firebase Auth's **signOut()** method, then redirects the user to the log in screen.

```
// Handle Logout button click
const logoutButton = document.getElementById('logoutBtn');
if (logoutButton) {
  logoutButton.addEventListener('click', async (event) => {
    event.preventDefault();
    try {
      await signOut(auth);
      console.log("User signed out successfully.");
      window.location.href = "/login.html";
    } catch (error) {
      console.error("Error signing out:", error);
    }
  });
}
```

Figure 87 – `logoutButton.addEventListener()` – JS

**`confirmDeleteBtn.addEventListener()`** When the user confirms account deletion in the modal, this function removes their data from both Firestore, Storage, and Firebase Auth. If the user session is expired it prompts the user to log in again.

```
if (confirmDeleteBtn) {
  confirmDeleteBtn.addEventListener("click", async () => {
    const user = auth.currentUser;

    if (user) {
      try {
        // Delete user's profile photo from Firebase Storage
        const storage = getStorage(app);
        const photoRef = ref(storage, `profilePhotos/${user.uid}.jpg`);

        try {
          await deleteObject(photoRef);
          console.log("Profile photo deleted from storage.");
        } catch (storageError) {
          console.warn("No profile photo found or error deleting it:", storageError.message);
        }

        // Delete Firestore user document
        await deleteDoc(doc(db, "users", user.uid));

        // Delete the authenticated user
        await user.delete();

        deleteStatusMessage.textContent = "Your account was successfully deleted. Redirecting...";
        const modal = bootstrap.Modal.getInstance(document.getElementById("deleteAccountModal"));
        modal.hide();
      }
    }
  });
}
```

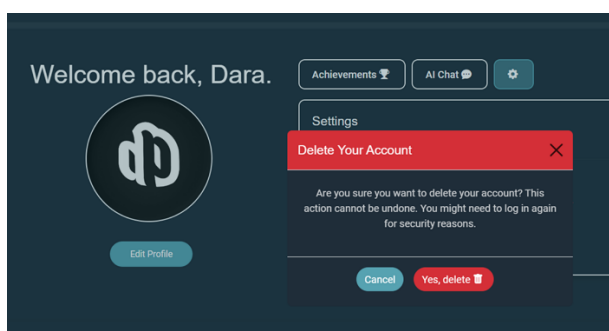
Figure 88 – `confirmDeleteBtn.addEventListener()` – JS

Figure 89 – Confirm Deletion UI

**displayLevelData()** This function displays data for level completion from Firestore. It shows - Loading.. then checks if the level is completed or not and displays the correct UI card, completed, or level not completed yet.

```
// Function to display level data from Firestore
function displayLevelData(levels) {
  const container = document.getElementById("levelsContainer");
  container.innerHTML = "";

  // Add loading state for levels in the card
  const loadingCard = document.createElement("div");
  loadingCard.className = "widget";
  loadingCard.innerHTML = `
    <h5>Loading Levels...</h5>
    <div class="text-center">
      <div class="spinner-border text-primary" role="status">
        <span class="visually-hidden">Loading...</span>
      </div>
    </div>
  `;
}
```

Figure 90 – displayLevelData() – JS

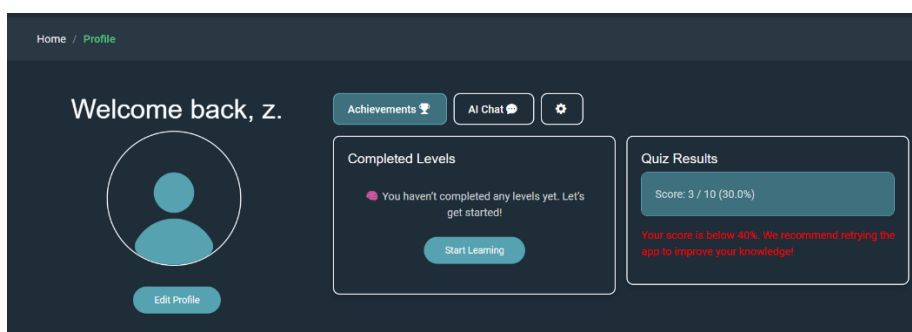


Figure 91 – User who took the quiz but didn't complete the level

#### 5.4.2.2 VR Application

##### 5.4.2.2.1 Firebase

To import Firebase into my project I had to find a version that worked with the version of Unity I was using, I found a couple on GitHub and used trial and error/Online forums until one finally worked.

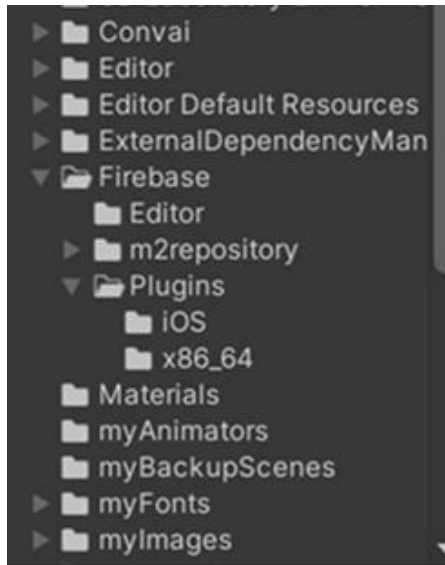


Figure 92 - Firebase Unity Plugin Import from GitHub

In C# it uses the **Awake()** method for setting it up. It then does a dependency check to ensure Firebase is ready before using it.

```

10
11 {
12     public static AuthManager Instance { get; private set; } // Singleton Instance
13
14     private FirebaseAuth auth;
15     private FirebaseUser user;
16     private Firestore db;
17
18     [Header("Scene Management")]
19     public SceneManager sceneManager; // Reference to SceneManager for scene transit
20
21     void Awake()
22     {
23         // Ensure only one instance exists
24         if (Instance == null)
25         {
26             Instance = this;
27             DontDestroyOnLoad(gameObject); // Keep it across scenes
28         }
29         else
30         {
31             Destroy(gameObject); // Destroy duplicate
32             return;
33         }
34
35         auth = FirebaseAuth.DefaultInstance; // Initialize Firebase Auth
36         db = Firestore.DefaultInstance; // Initialize Firestore
37     }
38
39     async void Start()
40     {
41         await InitializeFirebase();
42     }
43
44     private async Task InitializeFirebase()
45     {
46         var dependencyStatus = await FirebaseApp.CheckAndFixDependenciesAsync();
47         if (dependencyStatus == DependencyStatus.Available)
48         {
49             auth = FirebaseAuth.DefaultInstance;
50         }
51     }
52 }

```

Figure 93 – Initialising Firebase in Unity

The application isn't big, but I have eighteen scripts, and I can't document every method, so here's an example of a C sharp script method for updating the level completion.

The method at the bottom with **SetOptions.MergeAll** is updating Firestore when the user clicks the UI button on the menu in the 'Learn about the brain' scene to continue to the Quiz.

```
// Call this from a button in the level to mark it complete
public void MarkLevelComplete(string levelName)
{
    if (user == null)
    {
        Debug.LogError("User not logged in. Cannot update Firestore.");
        return;
    }

    DatabaseReference docRef = db.Collection("users").Document(user.UserId);

    // Create a nested dictionary for levelsCompleted
    Dictionary<string, object> levelUpdate = new Dictionary<string, object>
    {
        { "levelsCompleted", new Dictionary<string, object> { { levelName, true } } }
    };

    // Perform Firestore update to set the new level completion status
    docRef.SetAsync(levelUpdate, SetOptions.MergeAll).ContinueWith(task =>
    {
        if (task.IsCompleted)
        {

```

Figure 94 – Level Completed C Sharp Code

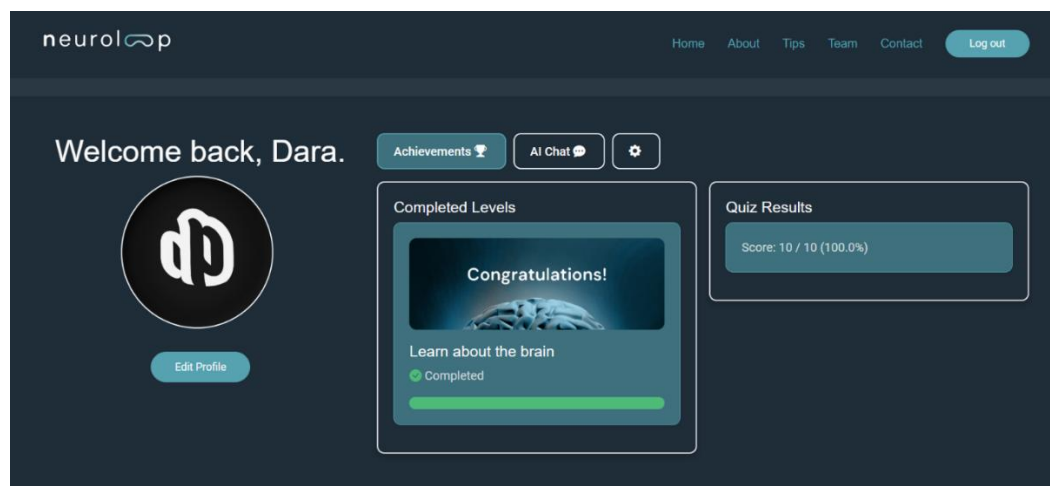


Figure 95 – Learn about the brain – Completed UI

#### 5.4.2.2.2 Convai API

This was not as straightforward as it may seem, it took a long time to understand how to make this work. I had to edit the Convai prefabs to match the character panel UI and make it similar to my own, I also had to search and edit the script to allow for the user to use a VR controller and map a button to the function listening for the user input.

Here is the Covai website where I obtained the API key.

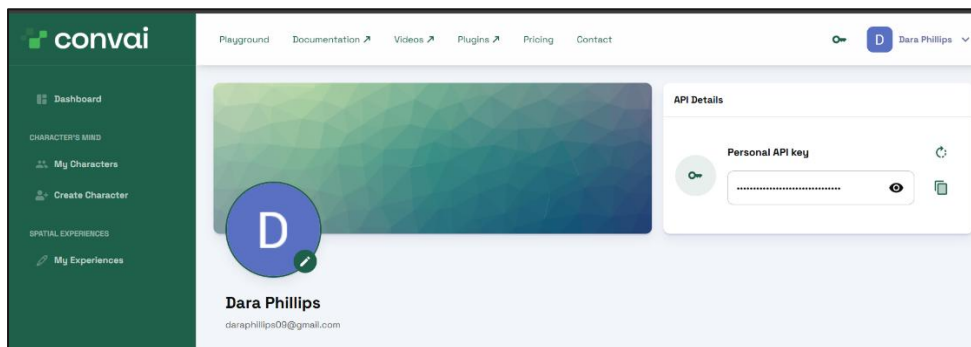


Figure 96 – Convai API Key

This is how I set up Nancy (The AI character).

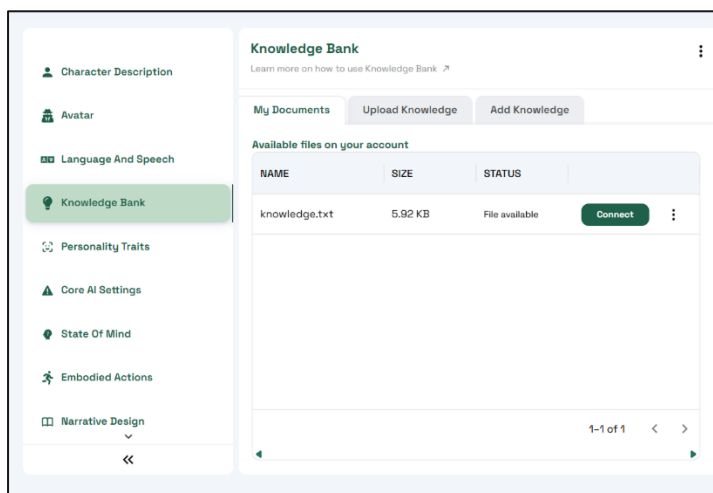


Figure 97 – Character setup

When I had the plugin imported into Unity, I update the key here. As you can there's 100 free interactions per month on the free plan I'm using.

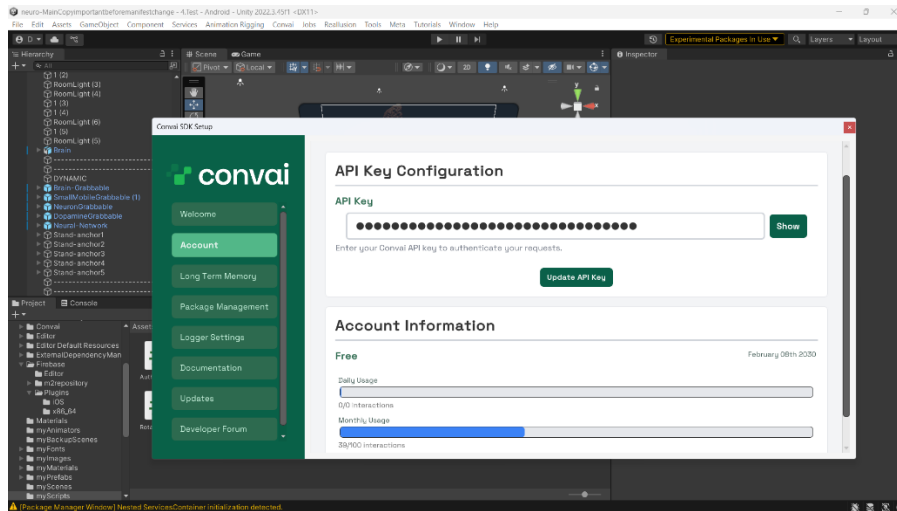


Figure 98 – Updating the API Key in Unity

The Ready Player Me editor.

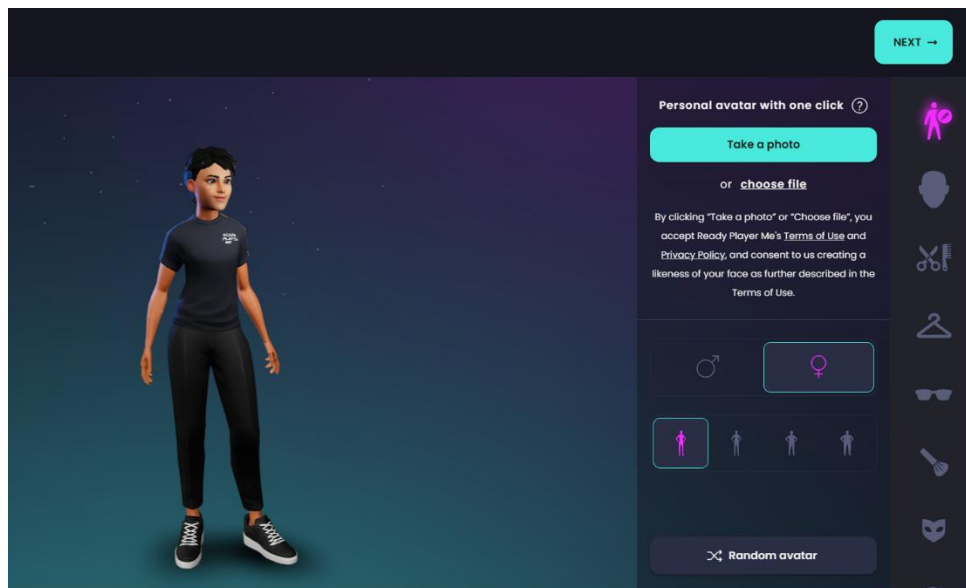


Figure 99 – Customising Character - Ready Player Me

## 6 User Testing

### 6.1 Changes made based on User Feedback

I tested throughout the project for both the website and the VR app. I used the patience of family and friends to discover issues I may have overlooked as I was developing. For the Web app, I conducted a Think Aloud and Heuristic Evaluation with two of my classmates and created a second google forms survey. I distributed the survey to my classmates nearing the end of development to gather qualitative and quantitative data and make any last-minute changes that I could with the time I had left. I asked them to complete four tasks and to fill out the form. For the VR application, I conducted a Think Aloud with one of my classmates and got help from a friend. I created a video of the game play in Blender and distributed it to friends who didn't have access to a Meta Quest headset to gain additional feedback – [Link to the YouTube video.](#)

#### 6.1.1 Web Application

##### 6.1.1.1 Classmate Survey - Tasks

1. Sign up.
2. Log in.
3. Read one of the pages from the cards on the 'Tips' section of the homepage.
4. Send feedback in the 'Contact' section on the homepage.

##### 6.1.1.2 Web App Changes

Based on the feedback I received from the Think Aloud and Heuristic Evaluation and made the following changes to the neuroloop website.

- Changed the delete profile modal header from green to red
- Ensured that all pages had the same navbar links and that they worked
- Made the homepage title casing uniform
- Tried my best to optimize the website for loading speed
- Added white space in the about section
- Changed the preview for profile images to a circle (it was square)

- Added feedback for deleting your profile (you may need to log in again)
- Changed the article section to 'Tips' and changed the card CSS to be more clear

### 6.1.1.3 Think Aloud/Heuristic Evaluation – Feedback

#### 6.1.1.3.1 Positive Feedback Examples

| 6. Problem / Positive                                                                                                                                                                | Relevant Heuristic | Severity rating |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------|
| When using the sign up and login forms the system instantly tells you if you are filling in a field incorrectly. This lets the user know immediately what is wrong and how to fix it | Error prevention   | Good            |

Figure 100 – Classmate 1

| 8. Problem / Positive                                                                                                                                                                    | Relevant Heuristic                | Severity rating |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------|
| I can navigate the website easily when viewing my profile thanks to the breadcrumb feature. Whenever I try to edit my profile, I can easily back out if I don't want to save any changes | Flexibility and efficiency of use | Good            |

Figure 101 – Classmate 2

#### 6.1.1.3.2 Negative Feedback Examples

| 4. Problem / Positive                                                                                                                                                                                                                                     | Relevant Heuristic        | Severity rating |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------|
| Website headings: I wish all the headings on the website all followed like the same grammatical rules and structures.<br>Example: "The Brain, Social Media, How to reclaim your time." and "How habits form". 1 title has all capitals, the other doesn't | Consistency and standards | Minor           |

Figure 102 – Classmate 1

| 3. Problem / Positive                                                                                                                                                                                               | Relevant Heuristic              | Severity rating |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------|
| The "Articles" section, the heading for each article link is hard to read on first appearance. Even though they become visible when the user hovers over them, it would be nice if they were visible from the start | Aesthetic and minimalist design | Major           |

Figure 103 - Classmate 2

#### 6.1.2 VR application

Based on the feedback I received from the Think Aloud and my friend's comments, I made the following changes to the neuroloop VR app.

- Made the audio in the zones shorter
- Updated Nancy's initial voice interaction script on the Convai website to direct the user to the menu
- Added sound to scene transitions
- Added a restart button for the quiz (A button to go back was too complicated)

### 6.1.2.1 Friends - Positive Feedback

1. "This could be used in schools, it's so clever and would empower so many kids (and adults) to be more mindful and knowledgeable about their social media usage.
2. "Looks really cool."
3. "That's awesome! Well done man."

## 6.2 Screenshots – User feedback / Survey Results

### 6.2.1 VR App Feedback

Feedback from a friend who play tested the VR APP.

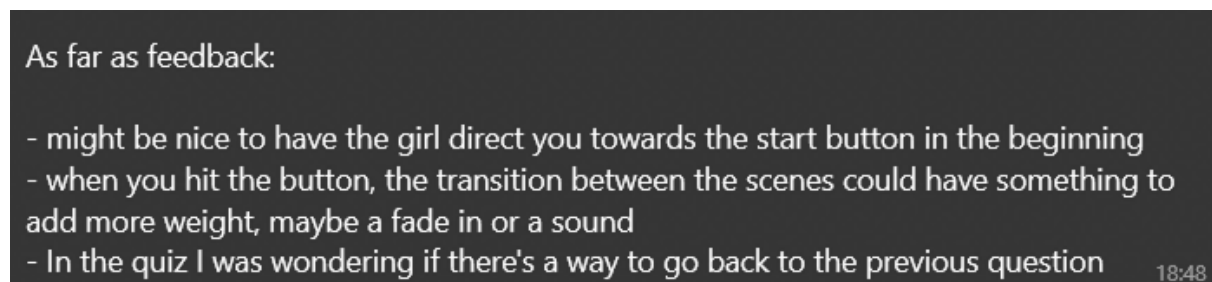


Figure 104 – VR App Feedback

### 6.2.2 Classmate Survey Results – Google Forms

Below are the results of my second survey for my website completed by my classmates.

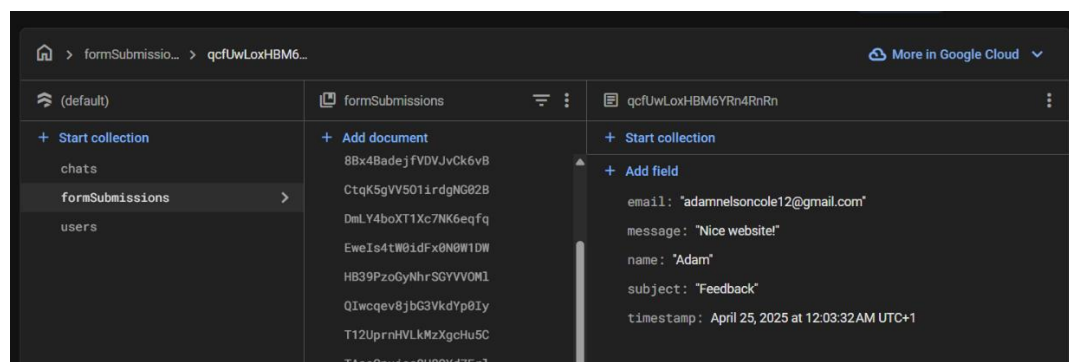


Figure 105 – Form Submission Feedback in Firestore – Adam

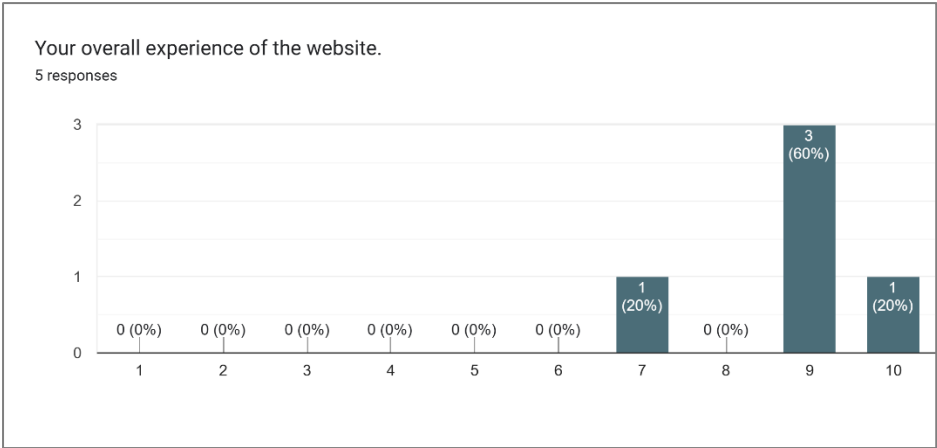


Figure 106 – Classmate Survey Q1

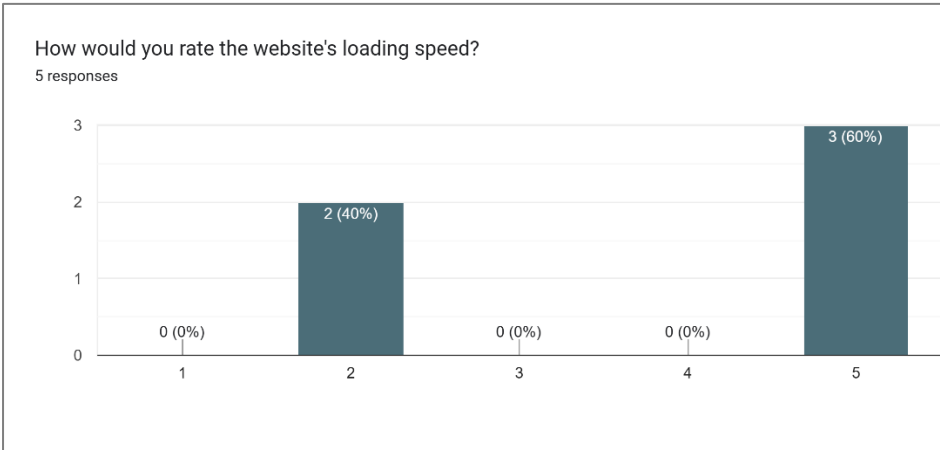


Figure 107 - Classmate Survey Q2

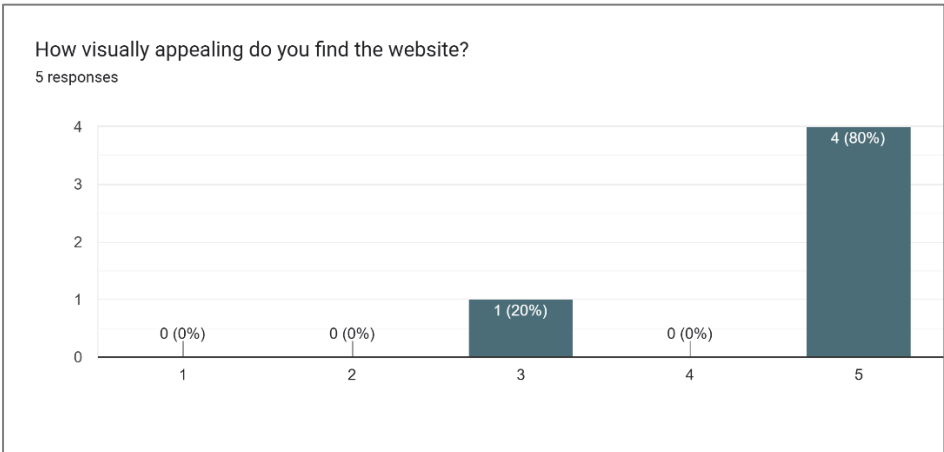


Figure 108 - Classmate Survey Q3

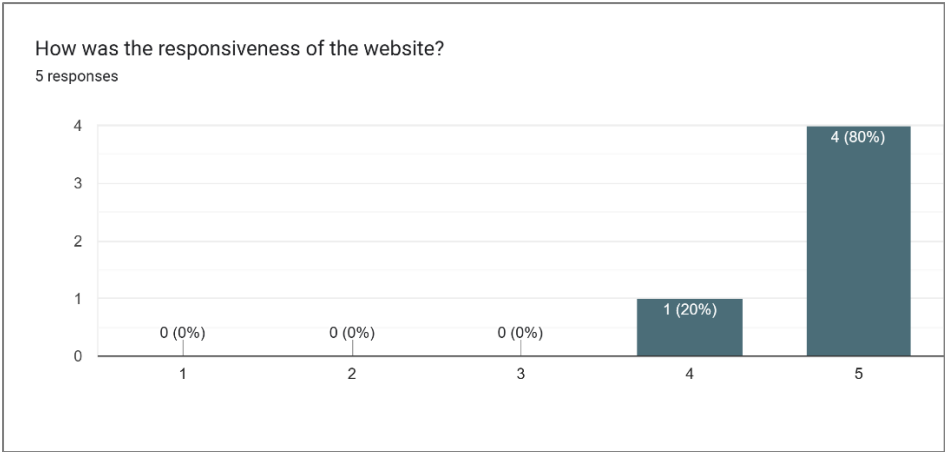


Figure 109 - Classmate Survey Q4

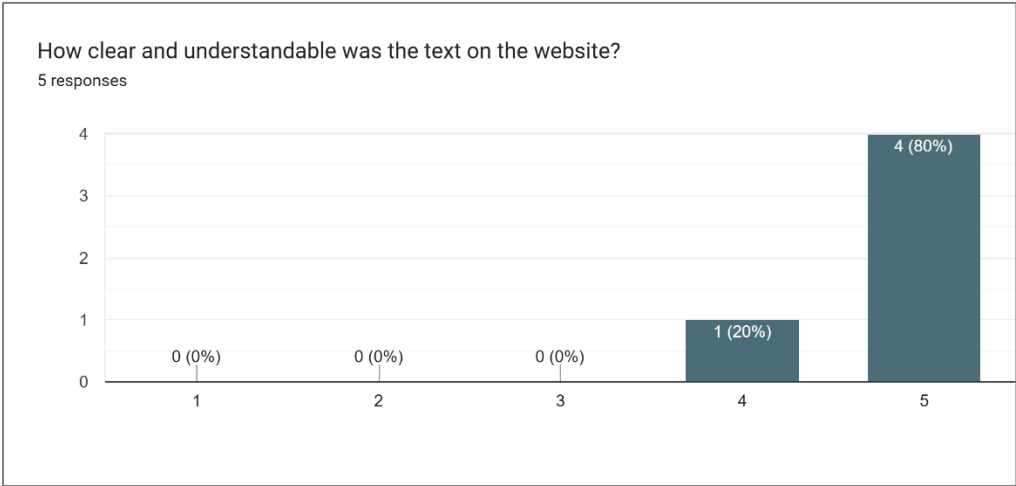


Figure 110 - Classmate Survey Q5

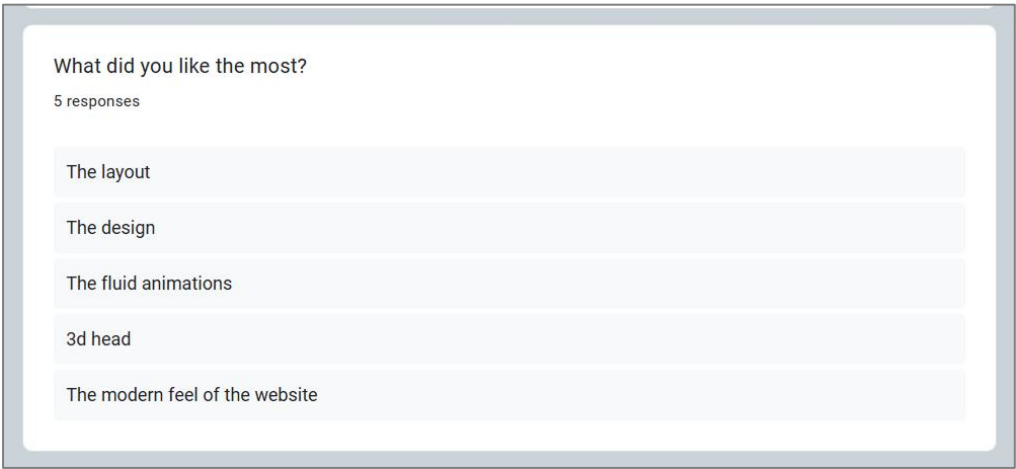


Figure 111 – Classmate Survey Q6

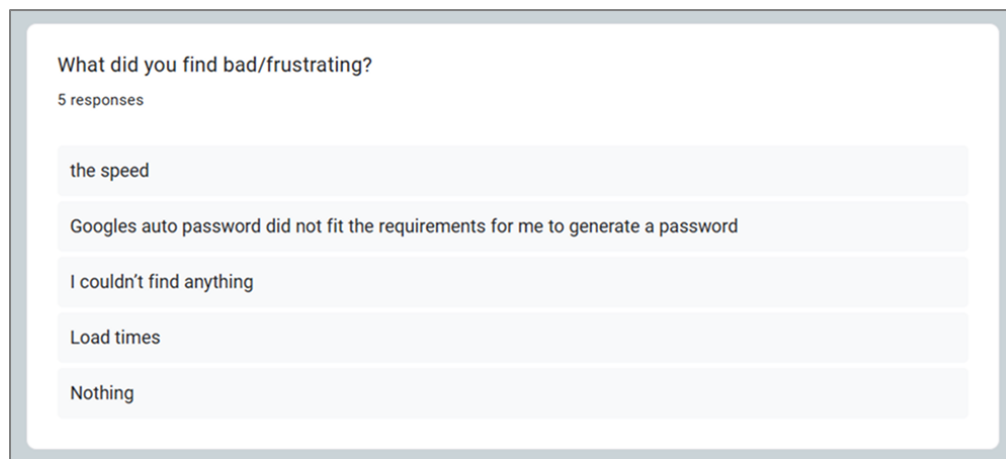


Figure 112 – Classmate Survey Q7

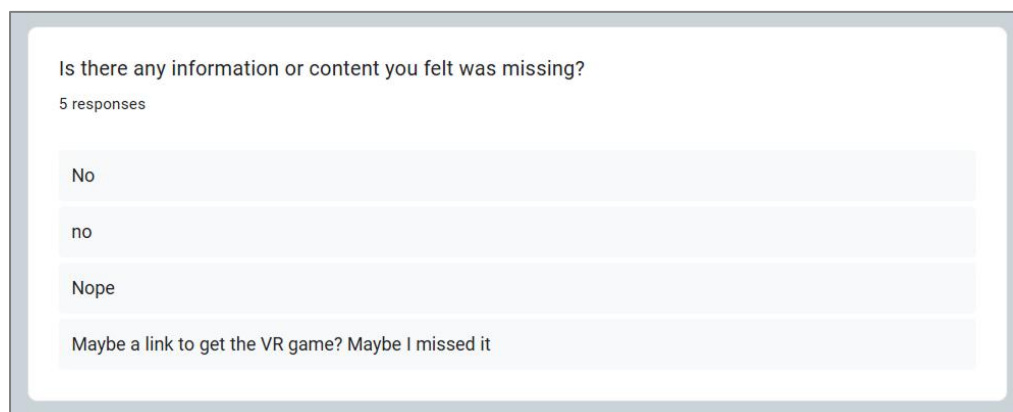


Figure 113 – Classmate Survey Q8

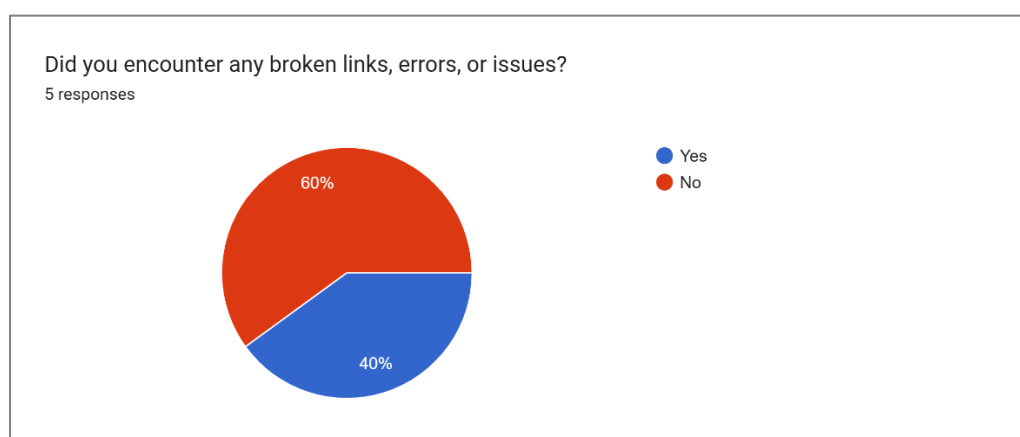
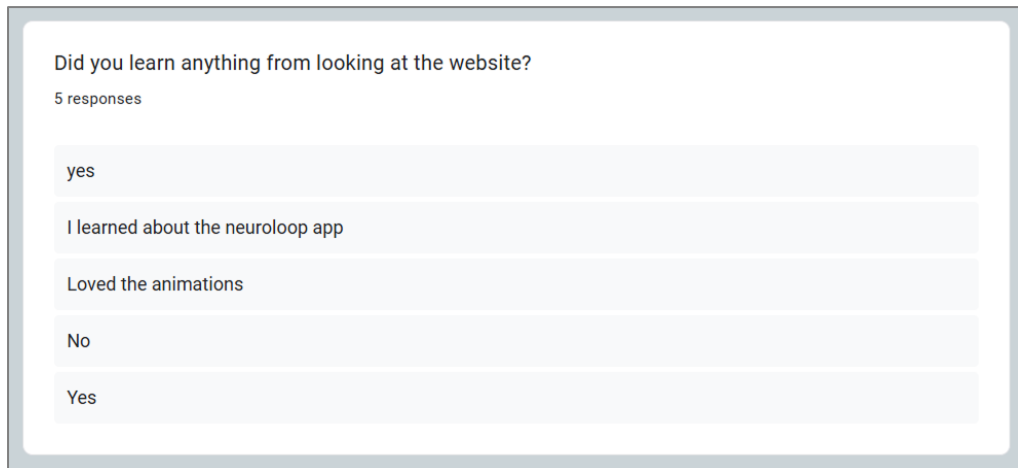


Figure 114 – Classmate Survey Q9



Did you learn anything from looking at the website?

5 responses

|                                   |
|-----------------------------------|
| yes                               |
| I learned about the neuroloop app |
| Loved the animations              |
| No                                |
| Yes                               |

Figure 115 – Classmate Survey Q10

## 7 Challenges & Reflection

### 7.1 Challenges

One of my main struggles originated when I chose to ignore advise from my supervisor John, in hindsight, I should have trusted his judgement. From the beginning of my VR app development I used Unity 6 (It was hard not to having attended the Unity conference) but this gave me a number of versioning and dependency issues when it came to integrating Firebase and uploading my application to the Meta Quest Developer Hub. This would allow me to back up my builds, publish the game to the store, and share the app with other developers/users. I can't express the trouble this caused me. My application is currently awaiting approval so it can be published. Unfortunately, the link on my website only leads to a 404 page I created and not to the Meta Store. I'm not sure if you've ever tried, but even sending it in for approval is a long and tedious process.

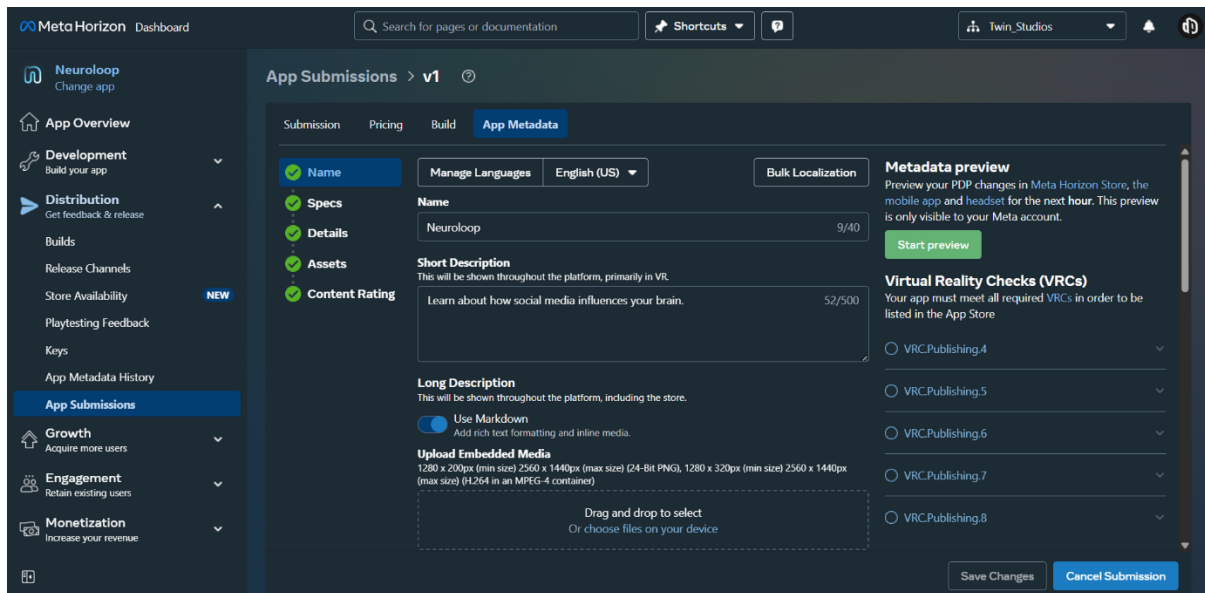


Figure 116 - VR App Submission

Another challenge I found quite difficult was optimization the performance in VR. For this reason I kept the amount of assets I used to a minimum and constantly had to assess what assets were to big or the amount lights I was using. I split the app into four scenes as a work around.

Haptic feedback was an element I desperately tried to implement into the VR app but unfortunately I couldn't figure it out and had to admit defeat. Even though the project has reached an end this is one aspect that really bugs me.

The topic I chose was definitely broad. The brain is complex, choosing the content and presenting it in a digestible way – especially to an audience that may not have studied science was certainly a challenge.

Unity was the hard part. Developing the website and integrating the API was a breeze in comparison. Despite these obstacles, the project has proven to be a valuable learning experience and I've definitely grown through the process. It pushed me to think critically, adapt quickly, and stay focused on the core objective: creating something that could inform and positively impact users.

## 7.2 Reflection

Developing Neuroloop has been an incredibly rewarding (and head banging) and eye-opening experience. From concept to completion, the project has allowed me to explore the intersection of immersive technology, neuroscience (albeit basic), and user behaviour.

This project strengthened my skills in Unity development, 3D modelling, UI/UX design, Web development, and storytelling. Working on Neuroloop also deepened my appreciation for how digital tools can be used not just for entertainment, but to support education, and self-awareness. If I was to undertake a project of this nature again, I would definitely use more play testers for the VR application. Limitations that held me back from gaining more user feedback include; uploading my builds to the Meta developer hub, fear of damaging the headset (the lenses are light-sensitive and finding safe space and time for testing was tricky). All in all, I feel the project was a success and I have a strong sense of accomplishment. I'm excited to demonstrate the work and possibly building on it further in the future.

## 8 Future Work

While I believe that Neuroloop successfully demonstrates the potential of immersive VR for neuroscience education and digital well-being, there are several opportunities to expand and enhance the project in the future.

### 8.1 Expanded Content and Zones

Future iterations could introduce additional brain regions, neurotransmitters, and behavioural psychology concepts. This would offer users a more complete understanding of how different systems – like memory, decision making, and stress interact with social media use.

### 8.2 Personalisation and Progress Tracking

Integrating users profile more effectively in the VR app, adding a system that adapts the content based on the quiz results or user interaction could create a deeper, more tailored experience.

### 8.3 Web Integration and Mobile Companion App

Although a responsive companion website currently exists, future work could involve developing a mobile app to extend the learning experience in new ways. This could include self-reflection, habit trackers, or even tailored content for schools and universities. Bringing Neuroloop into formal learning environments would be an exciting opportunity and one I'd love to explore.

### 8.4 Accessibility and Localisation

Future updates could also address accessibility needs (e.g., subtitles) and expand the experience to other languages to reach a broader audience.

## 9 Table of figures

|                                                                     |    |
|---------------------------------------------------------------------|----|
| Figure 1 – Survey q1. ....                                          | 5  |
| Figure 2 – Survey q2. ....                                          | 6  |
| Figure 3 – Survey q3. ....                                          | 6  |
| Figure 4 – Survey q4. ....                                          | 7  |
| Figure 5 – Research (Stsiampkouskaya, Joinson and Piwek, 2021)..... | 7  |
| Figure 6 - Survey q5.....                                           | 8  |
| Figure 7 - Survey q6.....                                           | 8  |
| Figure 8 - Survey q7.....                                           | 9  |
| Figure 9 - Survey q8.....                                           | 10 |
| Figure 10 - Survey q9. ....                                         | 10 |
| Figure 11 – Survey q10 .....                                        | 11 |
| Figure 12 – Persona 1 .....                                         | 12 |
| Figure 13 – Persona 2 .....                                         | 13 |
| Figure 14 – Word Cloud .....                                        | 14 |
| Figure 15 – headspace. ....                                         | 15 |
| Figure 16 - Tripp. ....                                             | 16 |
| Figure 17 - Moment.....                                             | 17 |
| Figure 18 – Research (Zolezzi et al., 2024). ....                   | 18 |
| Figure 19 – Figma logo.....                                         | 19 |
| Figure 20 – Unity logo.....                                         | 20 |
| Figure 21 – Firebase logo.....                                      | 20 |
| Figure 22 – VS Code logo. ....                                      | 21 |
| Figure 23 – Convai logo.....                                        | 21 |
| Figure 24 – Blender logo.....                                       | 22 |
| Figure 25 – Spline logo.....                                        | 22 |
| Figure 26 – Bootstrap logo.....                                     | 23 |
| Figure 27 – ChatGPT logo.....                                       | 23 |
| Figure 28 – GitHub logo .....                                       | 24 |
| Figure 29 – Meta Quest 3 .....                                      | 24 |
| Figure 30 – Realtime Colors.....                                    | 25 |
| Figure 31 – Chosen Colour Palette.....                              | 26 |
| Figure 32 – Accessibility Checking .....                            | 27 |
| Figure 33 – Typeface – DM Sans .....                                | 27 |
| Figure 34 – Original Plans.....                                     | 28 |
| Figure 35 – Example of flat UI icons .....                          | 28 |
| Figure 36 – Logo Moodboard.....                                     | 29 |
| Figure 37 – VR Moodboard.....                                       | 29 |
| Figure 38 – Web App Moodboard .....                                 | 29 |

|                                                                                                                                   |    |
|-----------------------------------------------------------------------------------------------------------------------------------|----|
| Figure 39 – Early Logo Iterations .....                                                                                           | 30 |
| Figure 40 – Chosen Logo .....                                                                                                     | 30 |
| Figure 41 – Final Logo .....                                                                                                      | 30 |
| Figure 42 – Sound Design / Audio Feedback .....                                                                                   | 31 |
| Figure 43 – Web App Architecture .....                                                                                            | 32 |
| Figure 44 - Mobile homepage; Above the fold, Articles, Habits, How it works?, Test Feedback, Contact us, Footer – wireframes..... | 33 |
| Figure 45 Sign up, T&C's, Login, Forgot Password – wireframes.....                                                                | 34 |
| Figure 46 Hamburger menu – wireframe .....                                                                                        | 34 |
| Figure 47 Read Article, All Articles, Profile, Edit Profile - wireframes .....                                                    | 35 |
| Figure 48 Desktop homepage – wireframes .....                                                                                     | 35 |
| Figure 49 Desktop homepage – wireframes .....                                                                                     | 36 |
| Figure 50 Desktop homepage – wireframes .....                                                                                     | 36 |
| Figure 51 – User/Guest Feedback.....                                                                                              | 37 |
| Figure 52 - Desktop Homepage - wireframe 1 .....                                                                                  | 38 |
| Figure 53 Desktop Homepage - wireframe 2 .....                                                                                    | 38 |
| Figure 54 Desktop Homepage - wireframe 3 .....                                                                                    | 39 |
| Figure 55 Desktop Homepage - wireframe 4 .....                                                                                    | 39 |
| Figure 56 Mobile Homepage - Mid-fi wireframes .....                                                                               | 40 |
| Figure 57 Mobile Signup/login with feedback messages - Mid-fi wireframes.....                                                     | 40 |
| Figure 58 Mobile Articles, Profile - Mid-fi wireframes.....                                                                       | 41 |
| Figure 59 Mobile hamburger menu - Mid-fi wireframes .....                                                                         | 41 |
| Figure 60 – VR App \ Architecture.....                                                                                            | 42 |
| Figure 61 – VR ‘Learn about your brain’ scene wireframe .....                                                                     | 42 |
| Figure 62 – Project Architecture.....                                                                                             | 43 |
| Figure 63 – Scene 1 .....                                                                                                         | 46 |
| Figure 64 Scene 2 – Instructions .....                                                                                            | 47 |
| Figure 65 – Corridor for anticipation.....                                                                                        | 47 |
| Figure 66 – Convai UI and Main Menu UI .....                                                                                      | 48 |
| Figure 67 – Learn about the brain - Zone 3.....                                                                                   | 48 |
| Figure 68 – Zone 1 .....                                                                                                          | 49 |
| Figure 69 – Quiz UI.....                                                                                                          | 49 |
| Figure 70 – Unity IDE – Scene Design .....                                                                                        | 50 |
| Figure 71 – Scene 2 – Blender Asset Example .....                                                                                 | 50 |
| Figure 72 – Accessible Image – Figma Import.....                                                                                  | 51 |
| Figure 73 – Learn about the brain – Navigation + Sounds .....                                                                     | 51 |
| Figure 74 – Audio Integration .....                                                                                               | 53 |
| Figure 75 – Initialising Firebase .....                                                                                           | 55 |
| Figure 76 - Firebase Authentication .....                                                                                         | 56 |
| Figure 77 – Firebase Functions Get Open AI Response .....                                                                         | 56 |

|                                                                       |    |
|-----------------------------------------------------------------------|----|
| Figure 78 – Firebase Functions getOpenAIResponse .....                | 57 |
| Figure 79 – ChatGPT-3.5 Turbo UI.....                                 | 57 |
| Figure 80 – Firestore Sign up Data.....                               | 58 |
| Figure 81 – Firestore Documents Console .....                         | 58 |
| Figure 82 – Firestore Edit profile Data .....                         | 59 |
| Figure 83 – Firestore Form Submission .....                           | 59 |
| Figure 84 – Firestore Storage Code .....                              | 60 |
| Figure 85 – Firebase Storage Console.....                             | 60 |
| Figure 86 – OnAuthStateChanged() - JS .....                           | 61 |
| Figure 87 – logoutButton.addEventListener() – JS .....                | 62 |
| Figure 88 – confirmDeleteBtn.addEventListener() – JS.....             | 62 |
| Figure 89 – Confirm Deletion UI .....                                 | 62 |
| Figure 90 – displayLevelData() – JS.....                              | 63 |
| Figure 91 – User who took the quiz but didn't complete the level..... | 63 |
| Figure 92 - Firebase Unity Plugin Import from GitHub.....             | 64 |
| Figure 93 – Initialising Firebase in Unity .....                      | 64 |
| Figure 94 – Level Completed C Sharp Code.....                         | 65 |
| Figure 95 – Learn about the brain – Completed UI .....                | 65 |
| Figure 96 – Convai API Key .....                                      | 66 |
| Figure 97 – Character setup.....                                      | 66 |
| Figure 98 – Updating the API Key in Unity.....                        | 67 |
| Figure 99 – Customising Character - Ready Player Me.....              | 67 |
| Figure 100 – Classmate 1 .....                                        | 69 |
| Figure 101 – Classmate 2 .....                                        | 69 |
| Figure 102 – Classmate 1 .....                                        | 70 |
| Figure 103 - Classmate 2.....                                         | 70 |
| Figure 104 – VR App Feedback.....                                     | 71 |
| Figure 105 – Form Submission Feedback in Firestore – Adam.....        | 71 |
| Figure 106 – Classmate Survey Q1 .....                                | 72 |
| Figure 107 - Classmate Survey Q2 .....                                | 72 |
| Figure 108 - Classmate Survey Q3 .....                                | 72 |
| Figure 109 - Classmate Survey Q4 .....                                | 73 |
| Figure 110 - Classmate Survey Q5 .....                                | 73 |
| Figure 111 – Classmate Survey Q6.....                                 | 73 |
| Figure 112 – Classmate Survey Q7 .....                                | 74 |
| Figure 113 – Classmate Survey Q8.....                                 | 74 |
| Figure 114 – Classmate Survey Q9.....                                 | 74 |
| Figure 115 – Classmate Survey Q10 .....                               | 75 |
| Figure 116 - VR App Submission.....                                   | 76 |

## 10 References

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